

Geometry-based Methods in Vision

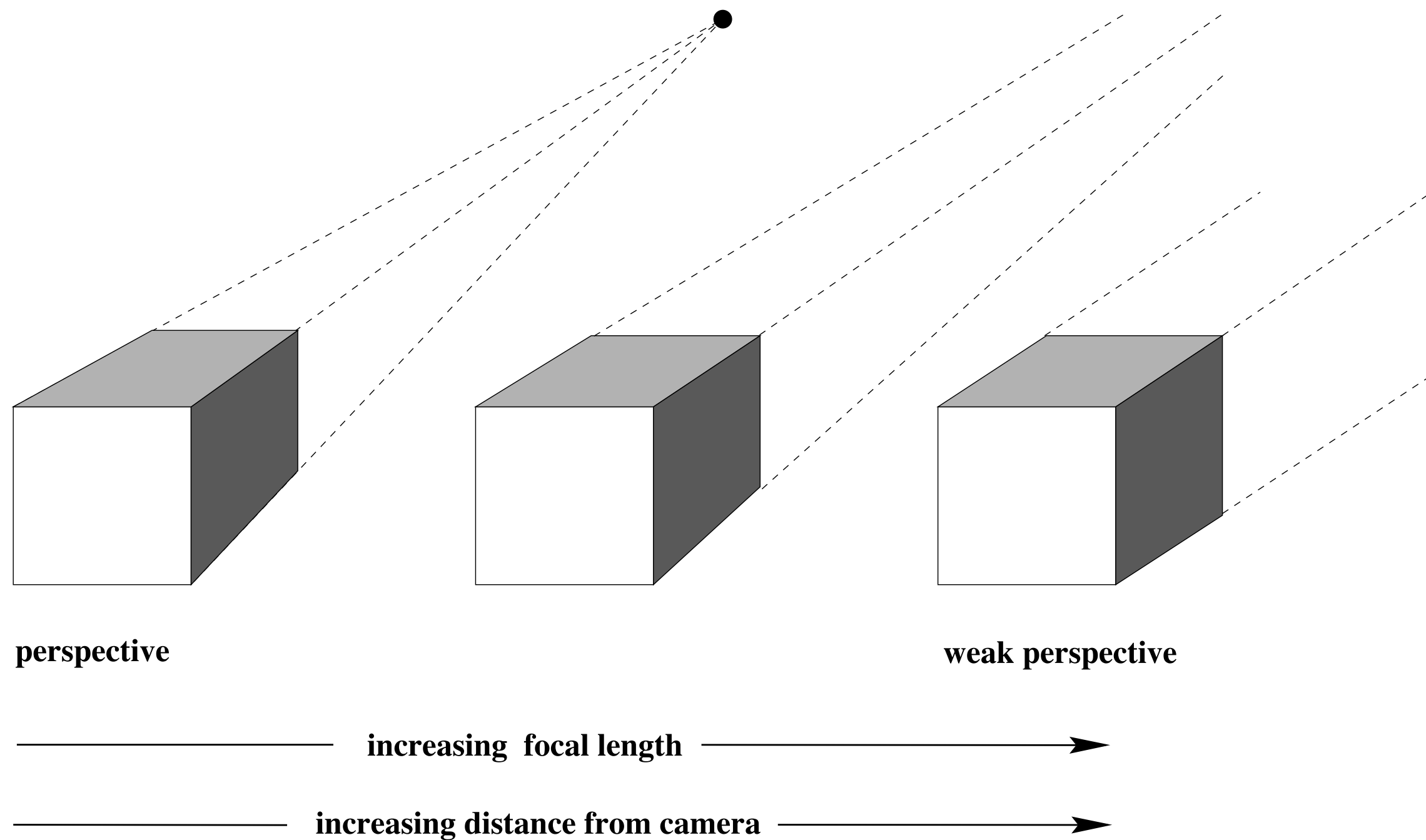
CMU 16-822, Fall 2024

Instructor: Shubham Tulsiani

Administrivia

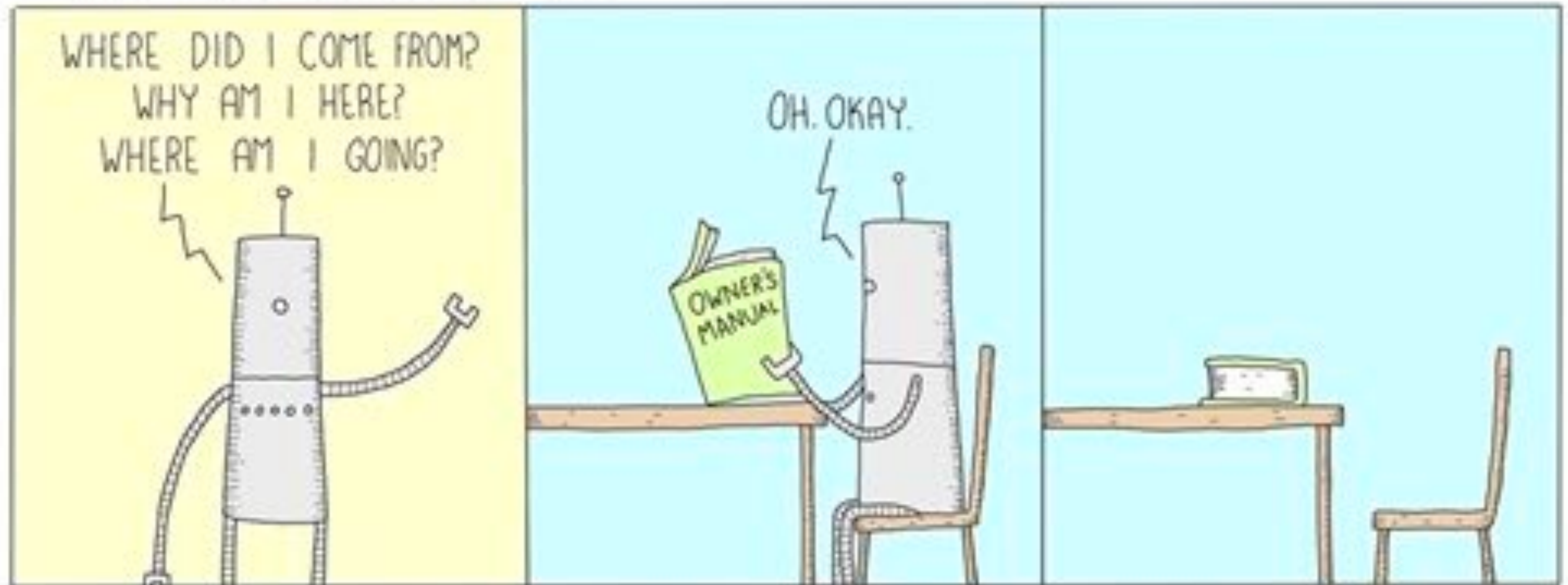
- PS1 out
 - Due **today**
 - *Reminder to self-report attendance*
 - *Please adhere to the template format*
- Assignment 1
 - *Due on Tue next week*

Lecture 7: Affine Cameras and Perspective-n-Points



A reminder to ask questions!

ROBOTS HAVE QUESTIONS TOO



Reminder: Camera Matrix

$$\mathbf{P} = \mathbf{K}R \begin{bmatrix} I & -\tilde{\mathbf{C}} \end{bmatrix}$$

Geometrically motivated view,
disentangling extrinsics and intrinsics

$$\mathbf{P} = \begin{bmatrix} \mathbf{M} & \mathbf{p}_4 \end{bmatrix}$$

A general projection from 3D to 2D

$$\text{rank}(\mathbf{P}) = 3$$

**Camera
Centre**

$$\mathbf{C} \text{ s.t. } \mathbf{P}\mathbf{C} = \mathbf{0}$$

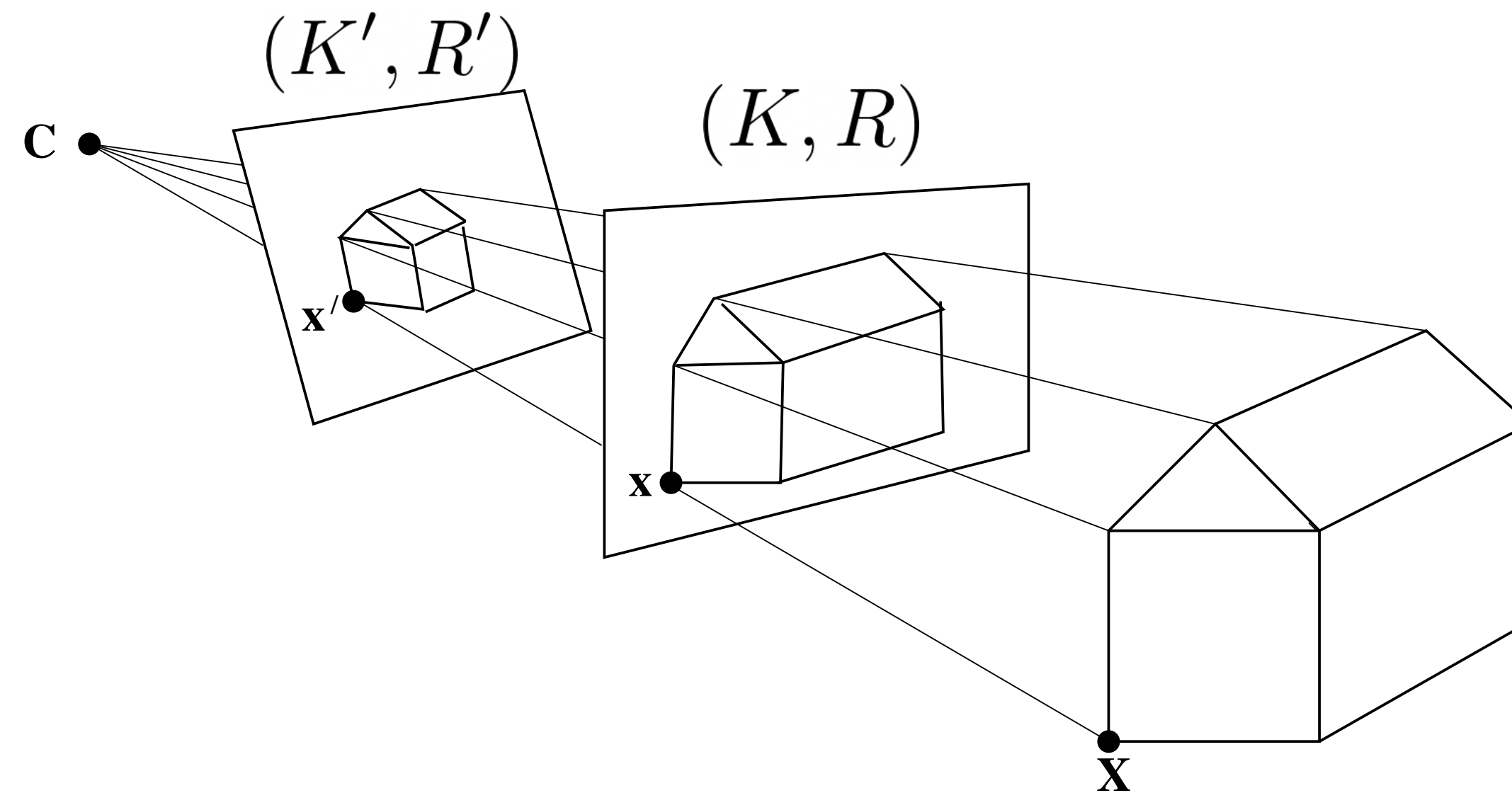
**Principal
Plane**

$$\pi \equiv \mathbf{p}^3$$

Unprojection

$$\lambda \mathbf{P}^+ \mathbf{u} + \mathbf{C}$$

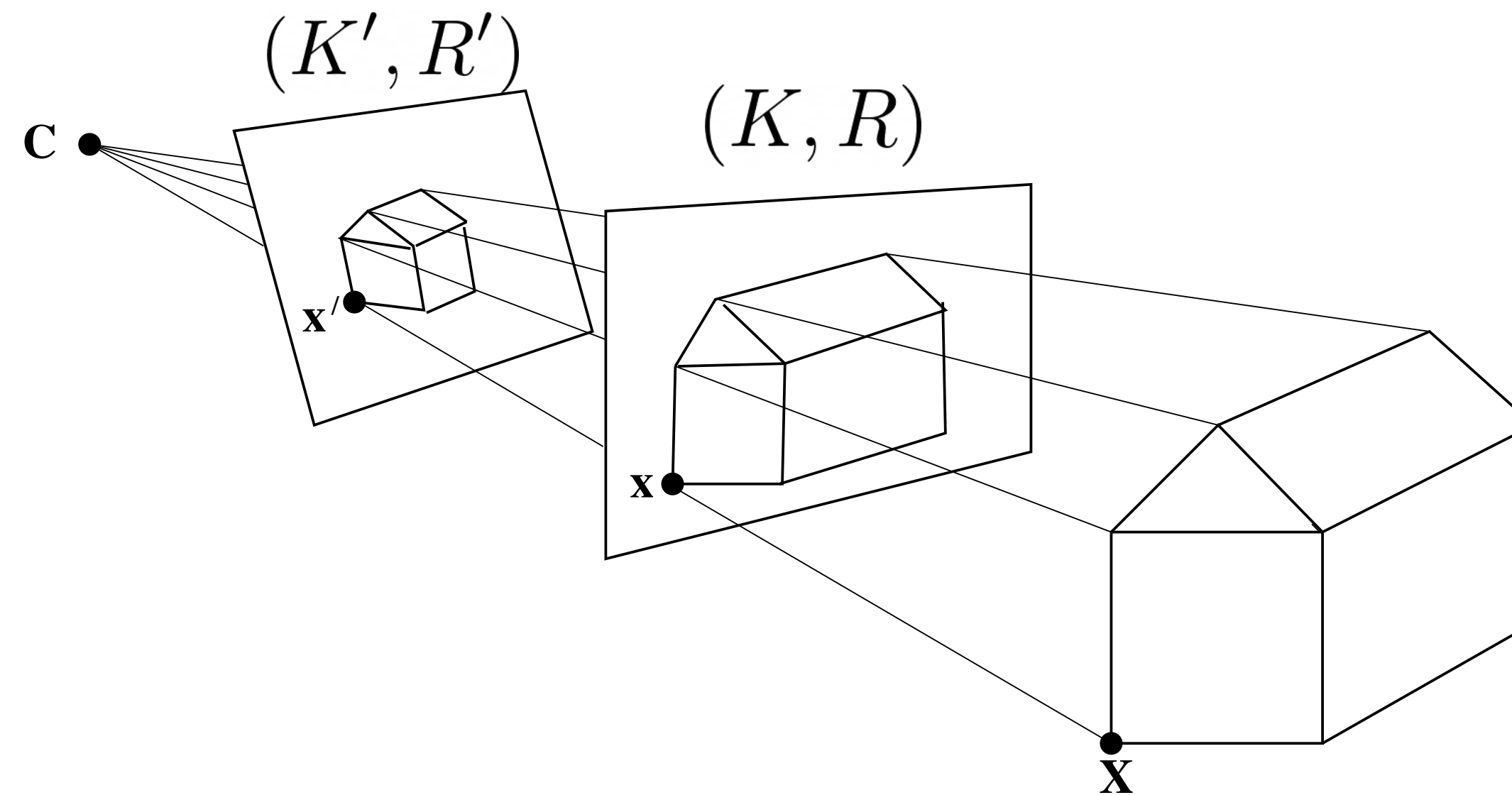
Relating Images with Fixed Camera Centre



Q: What is the type of transform relating x and x' ?

A 2D Homography! (one-to-one and preserves linearity)

Relating Images with Fixed Camera Centre

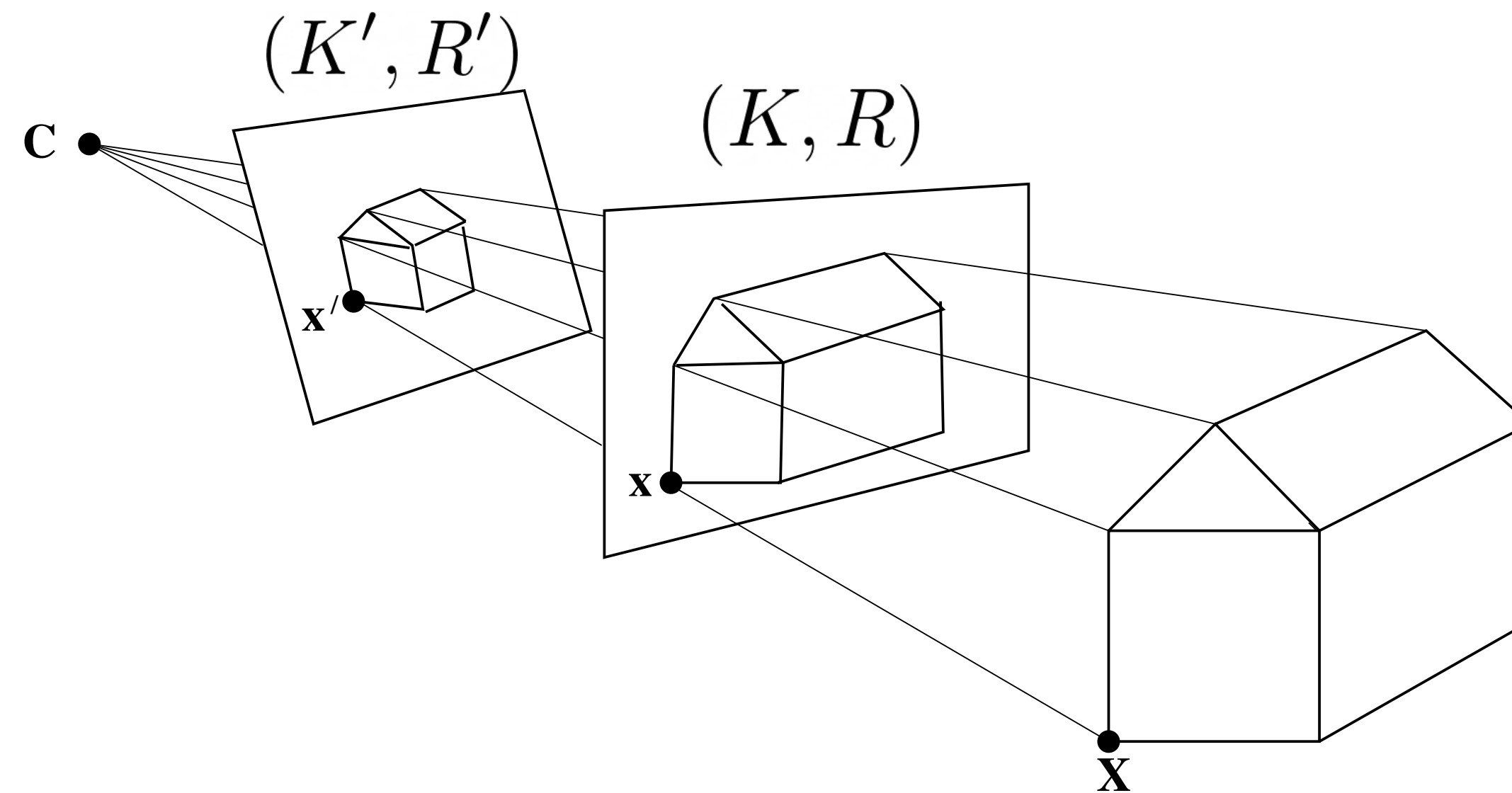


Q: What is the transform relating $\mathbf{x}$ and $\mathbf{x}'$?

$$\mathbf{x}' = \underbrace{K' R' R^{-1} K^{-1}}_{\mathbf{H}} \mathbf{x}$$

$$\begin{aligned} \mathbf{x}' &= \mathbf{P}' \mathbf{X} \\ &= K' R' [I \quad \tilde{\mathbf{C}}] \mathbf{X} \\ &= K' R' (K R)^{-1} \mathbf{P} \mathbf{X} \\ &= K' R' (K R)^{-1} \mathbf{x} \end{aligned}$$

Relating Images with Fixed Camera Centre



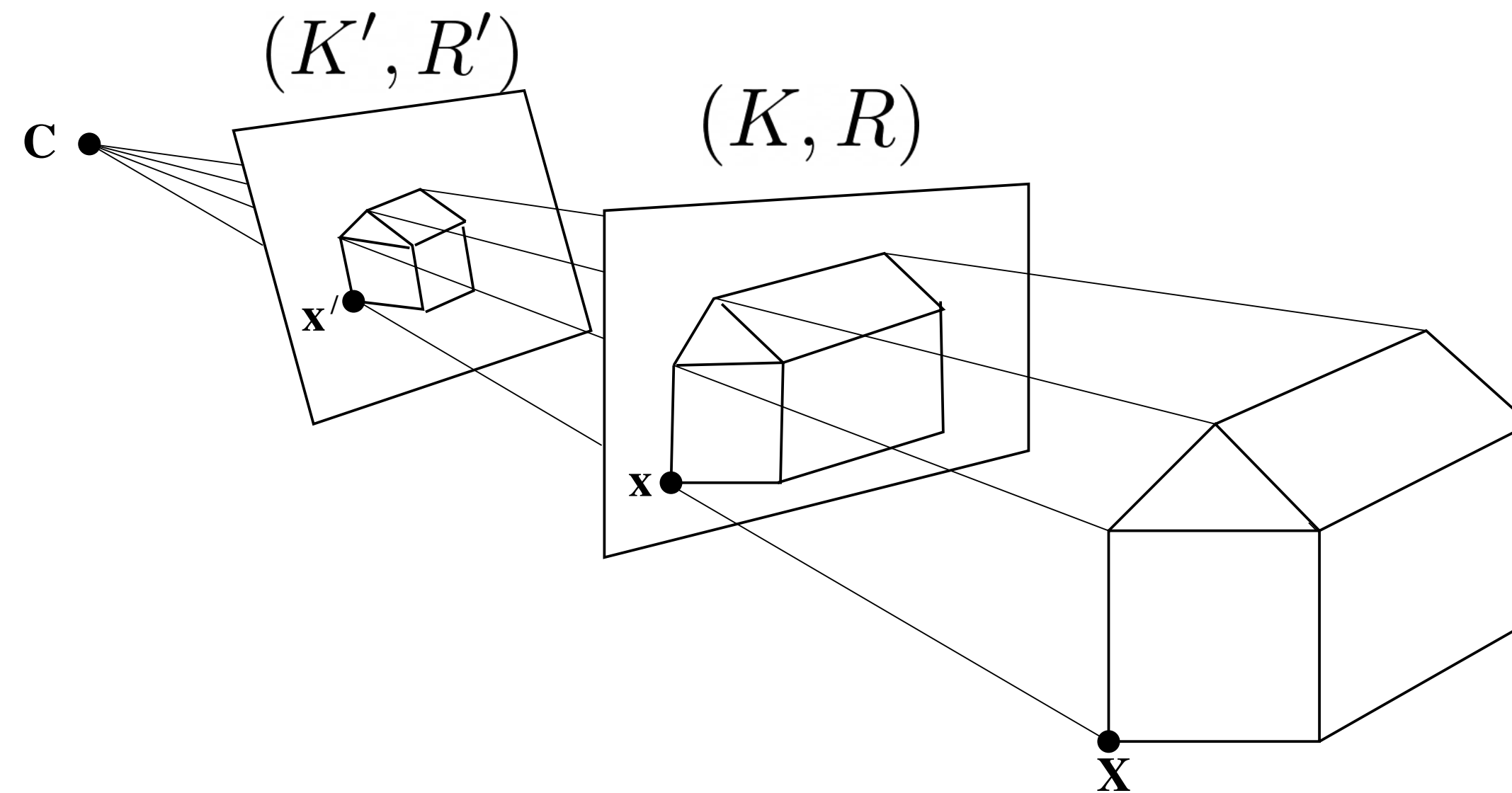
Q: What is the transform relating $\mathbf{x}$ and $\mathbf{x}'$ under zoom?

$\tilde{\mathbf{x}}' = k\tilde{\mathbf{x}} + (1 - k)\tilde{\mathbf{x}}_0$ distance from principal point is magnified by k

$$\mathbf{x}' = K' K^{-1} \mathbf{x}$$

$$K' = K \begin{bmatrix} k\mathbf{I} & \\ & 1 \end{bmatrix}$$

Relating Images with Fixed Camera Centre



Q: What is the transform relating $\mathbf{x}$ and $\mathbf{x}'$ if intrinsics are preserved?

$$\mathbf{x}' = K R K^{-1} \mathbf{x}$$

(assuming a **relative** rotation R between views)

Relating Images with Fixed Camera Centre



$$\mathbf{x}' = K R K^{-1} \mathbf{x}$$



The reason why panorama stitching works!

Cameras at Infinity

$$\mathbf{P} = [\mathbf{M} \quad \mathbf{p}_4]$$

$$\text{rank}(\mathbf{M}) = 2$$

So there exists $\mathbf{d}$ $\mathbf{M}\mathbf{d} = \mathbf{0}$

$$\mathbf{P} \begin{bmatrix} \mathbf{d} \\ 0 \end{bmatrix} = \mathbf{0} \quad \mathbf{C} \equiv \begin{bmatrix} \mathbf{d} \\ 0 \end{bmatrix}$$

**Affine
Cameras**

$$\mathbf{m}^{3\top} = \mathbf{0}$$

$$\mathbf{P} = \begin{bmatrix} \mathbf{m}^{1\top} & \mathbf{p}_4^1 \\ \mathbf{m}^{2\top} & \mathbf{p}_4^2 \\ \mathbf{0} & \mathbf{p}_4^3 \end{bmatrix} = \begin{bmatrix} \mathbf{m}^{1\top} & \mathbf{p}_4^1 \\ \mathbf{m}^{2\top} & \mathbf{p}_4^2 \\ \mathbf{0} & 1 \end{bmatrix}$$

Q: Why does the last element have to be non-zero for a valid $\mathbf{P}$?

Affine Cameras

$$\mathbf{P} = [\mathbf{M} \quad \mathbf{p}_4]$$

$$\text{rank}(\mathbf{M}) = 2$$

So there exists $\mathbf{d}$ $\mathbf{M}\mathbf{d} = \mathbf{0}$

$$\mathbf{P} \begin{bmatrix} \mathbf{d} \\ 0 \end{bmatrix} = \mathbf{0} \quad \mathbf{C} \equiv \begin{bmatrix} \mathbf{d} \\ 0 \end{bmatrix}$$

Affine
Cameras

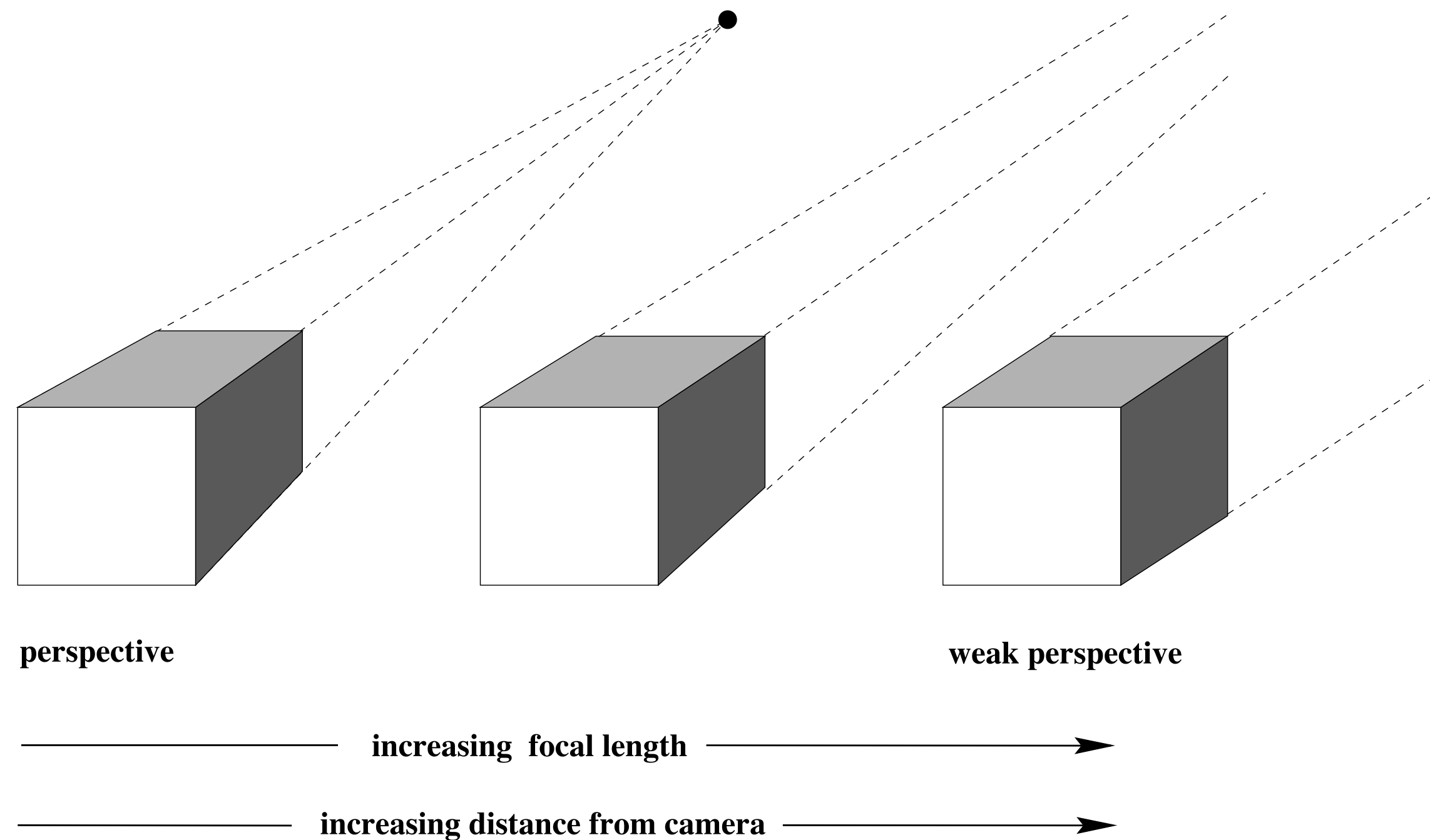
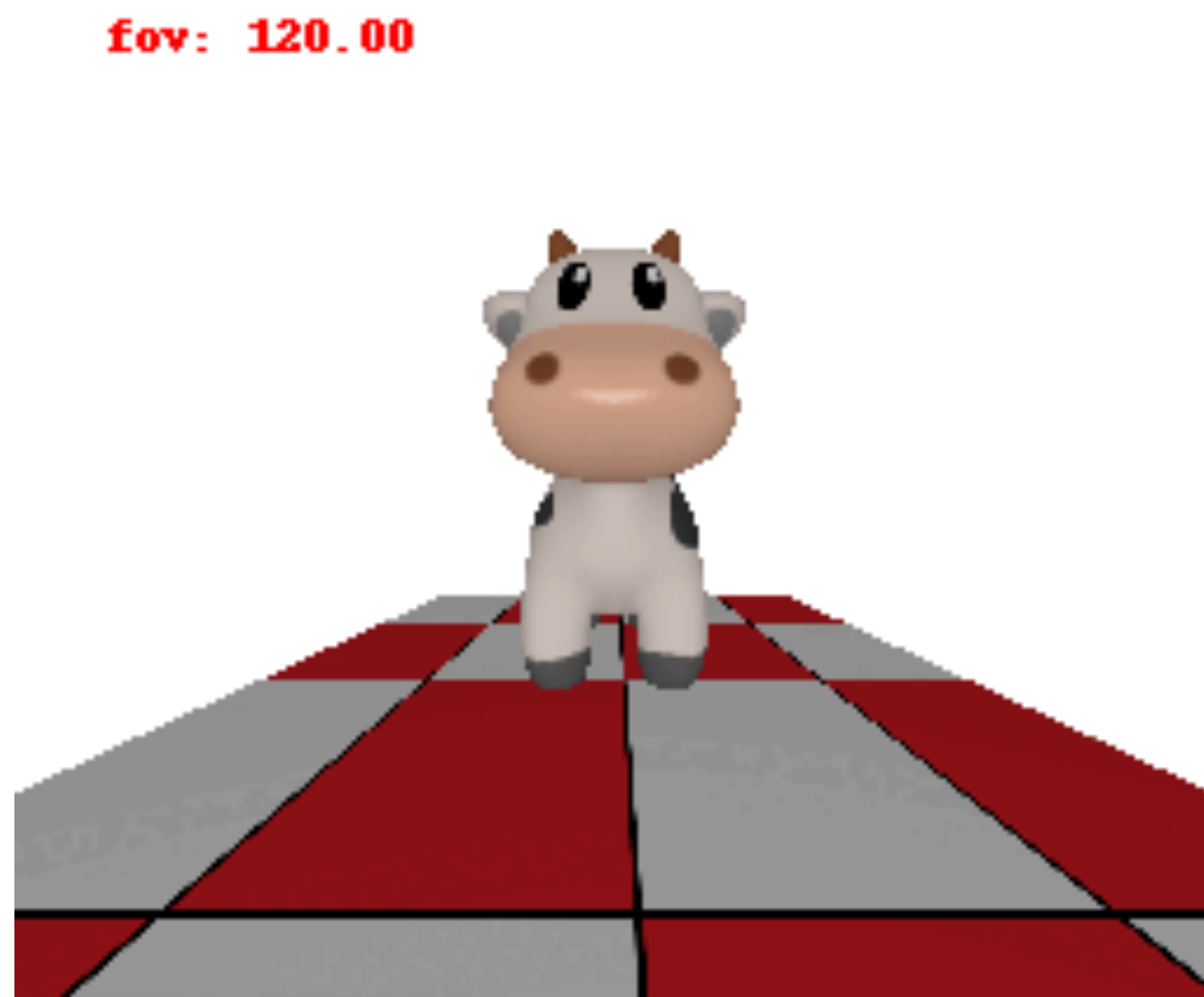
$$\mathbf{m}^{3\top} = \mathbf{0}$$

$$\mathbf{P} = \begin{bmatrix} \mathbf{m}^{1\top} & \mathbf{p}_4^1 \\ \mathbf{m}^{2\top} & \mathbf{p}_4^2 \\ \mathbf{0} & \mathbf{p}_4^3 \end{bmatrix} = \begin{bmatrix} \mathbf{m}^{1\top} & \mathbf{p}_4^1 \\ \mathbf{m}^{2\top} & \mathbf{p}_4^2 \\ \mathbf{0} & 1 \end{bmatrix}$$

$$x = \mathbf{m}^{1\top} \tilde{\mathbf{X}} + \mathbf{p}_4^1 \quad y = \mathbf{m}^{2\top} \tilde{\mathbf{X}} + \mathbf{p}_4^2$$

Image coordinates are an affine transform of the 3D coordinates!

Affine Cameras



Affine camera: infinitely far away,
and infinitely zoomed in

Affine Cameras

$$P_0 = KR[I \mid -\tilde{C}] = K \begin{bmatrix} \mathbf{r}^{1\top} & -\mathbf{r}^{1\top}\tilde{C} \\ \mathbf{r}^{2\top} & -\mathbf{r}^{2\top}\tilde{C} \\ \mathbf{r}^{3\top} & -\mathbf{r}^{3\top}\tilde{C} \end{bmatrix}$$

camera moving along 3rd axis with unit speed



$$\tilde{C}_t = \tilde{C} - t\mathbf{r}^3$$

$$P_t = K \begin{bmatrix} \mathbf{r}^{1\top} & -\mathbf{r}^{1\top}(\tilde{C} - t\mathbf{r}^3) \\ \mathbf{r}^{2\top} & -\mathbf{r}^{2\top}(\tilde{C} - t\mathbf{r}^3) \\ \mathbf{r}^{3\top} & -\mathbf{r}^{3\top}(\tilde{C} - t\mathbf{r}^3) \end{bmatrix} = K \begin{bmatrix} \mathbf{r}^{1\top} & -\mathbf{r}^{1\top}\tilde{C} \\ \mathbf{r}^{2\top} & -\mathbf{r}^{2\top}\tilde{C} \\ \mathbf{r}^{3\top} & d_t \end{bmatrix}$$

+ corresponding zoom



$$K_t = K \begin{bmatrix} d_t/d_0 & & \\ & d_t/d_0 & \\ & & 1 \end{bmatrix}$$

$$P_t = K \begin{bmatrix} d_t/d_0 & & \\ & d_t/d_0 & \\ & & 1 \end{bmatrix} \begin{bmatrix} \mathbf{r}^{1\top} & -\mathbf{r}^{1\top}\tilde{C} \\ \mathbf{r}^{2\top} & -\mathbf{r}^{2\top}\tilde{C} \\ \mathbf{r}^{3\top} & d_t \end{bmatrix} = \frac{d_t}{d_0} K \begin{bmatrix} \mathbf{r}^{1\top} & -\mathbf{r}^{1\top}\tilde{C} \\ \mathbf{r}^{2\top} & -\mathbf{r}^{2\top}\tilde{C} \\ \mathbf{r}^{3\top}d_0/d_t & d_0 \end{bmatrix}$$

$$P_\infty = \lim_{t \rightarrow \infty} P_t = K \begin{bmatrix} \mathbf{r}^{1\top} & -\mathbf{r}^{1\top}\tilde{C} \\ \mathbf{r}^{2\top} & -\mathbf{r}^{2\top}\tilde{C} \\ \mathbf{0}^\top & d_0 \end{bmatrix}$$

Affine Cameras

$$K \begin{bmatrix} \mathbf{r}^{1\top} & -\mathbf{r}^{1\top} \tilde{\mathbf{C}} \\ \mathbf{r}^{2\top} & -\mathbf{r}^{2\top} \tilde{\mathbf{C}} \\ \mathbf{0}^\top & d_0 \end{bmatrix} = \begin{bmatrix} K_{2 \times 2} & \tilde{\mathbf{x}}_0 \\ \hat{\mathbf{0}}^\top & 1 \end{bmatrix} \begin{bmatrix} \hat{\mathbf{R}} & \hat{\mathbf{t}} \\ \mathbf{0}^\top & d_0 \end{bmatrix}$$

2×1 2×3 2×1
 ~~2×1~~ ~~2×3~~ ~~2×1~~

Q: How many degrees of freedom?

$$3 + 2 + 3 + 2 + 1 = 11?$$

$$\mathbf{P} = \begin{bmatrix} K_{2 \times 2} \hat{R} & K_{2 \times 2} \hat{\mathbf{t}} + d_0 \tilde{\mathbf{x}}_0 \\ \mathbf{0} & d_0 \end{bmatrix}$$

Observation 1: Scaling d_0 and $K_{2 \times 2}$ by the same factor does not change $\mathbf{P}$!

Affine Cameras

$$K \begin{bmatrix} \mathbf{r}^{1\top} & -\mathbf{r}^{1\top} \tilde{\mathbf{C}} \\ \mathbf{r}^{2\top} & -\mathbf{r}^{2\top} \tilde{\mathbf{C}} \\ \mathbf{0}^\top & d_0 \end{bmatrix} = \begin{bmatrix} K_{2 \times 2} & \tilde{\mathbf{x}}_0 \\ \hat{\mathbf{0}}^\top & 1 \end{bmatrix} \begin{bmatrix} \hat{\mathbf{R}} & \hat{\mathbf{t}} \\ \mathbf{0}^\top & 1 \end{bmatrix}$$

2×1 2×3 2×1
 ~~2×1~~ ~~2×3~~ ~~2×1~~

Q: How many degrees of freedom?

$$3 + 2 + 3 + 2 = 10?$$

$$\mathbf{P} = \begin{bmatrix} K_{2 \times 2} \hat{R} & K_{2 \times 2} \hat{\mathbf{t}} + \tilde{\mathbf{x}}_0 \\ \mathbf{0} & 1 \end{bmatrix}$$

Observation 2: $\mathbf{t}$ and $\mathbf{x}_0$ are redundant

Affine Cameras

$$\mathbf{P} = \begin{bmatrix} \mathbf{K}_{2 \times 2} & \mathbf{0} \\ \hat{\mathbf{o}}^\top & 1 \end{bmatrix} \begin{bmatrix} \hat{\mathbf{R}} & \hat{\mathbf{t}} \\ \mathbf{0}^\top & 1 \end{bmatrix} = \begin{bmatrix} \mathbf{K}_{2 \times 2} & \tilde{\mathbf{x}}_0 \\ \hat{\mathbf{o}}^\top & 1 \end{bmatrix} \begin{bmatrix} \hat{\mathbf{R}} & \mathbf{0} \\ \mathbf{0}^\top & 1 \end{bmatrix}$$

Equivalent parametrizations

Left one is a more common in code, but remember that $\mathbf{t}$ is no longer reflective of 3D ‘position’ of camera!

(rather, it indicates 2D pixel coordinate where world origin projects)

Hierarchy of Affine Cameras

$$\mathbf{P} = \begin{bmatrix} \mathbf{K}_{2 \times 2} & \mathbf{0} \\ \hat{\mathbf{0}}^\top & 1 \end{bmatrix} \begin{bmatrix} \hat{\mathbf{R}} & \hat{\mathbf{t}} \\ \mathbf{0}^\top & 1 \end{bmatrix} = \begin{bmatrix} \mathbf{K}_{2 \times 2} & \tilde{\mathbf{x}}_0 \\ \hat{\mathbf{0}}^\top & 1 \end{bmatrix} \begin{bmatrix} \hat{\mathbf{R}} & \mathbf{0} \\ \mathbf{0}^\top & 1 \end{bmatrix}$$

Orthographic:

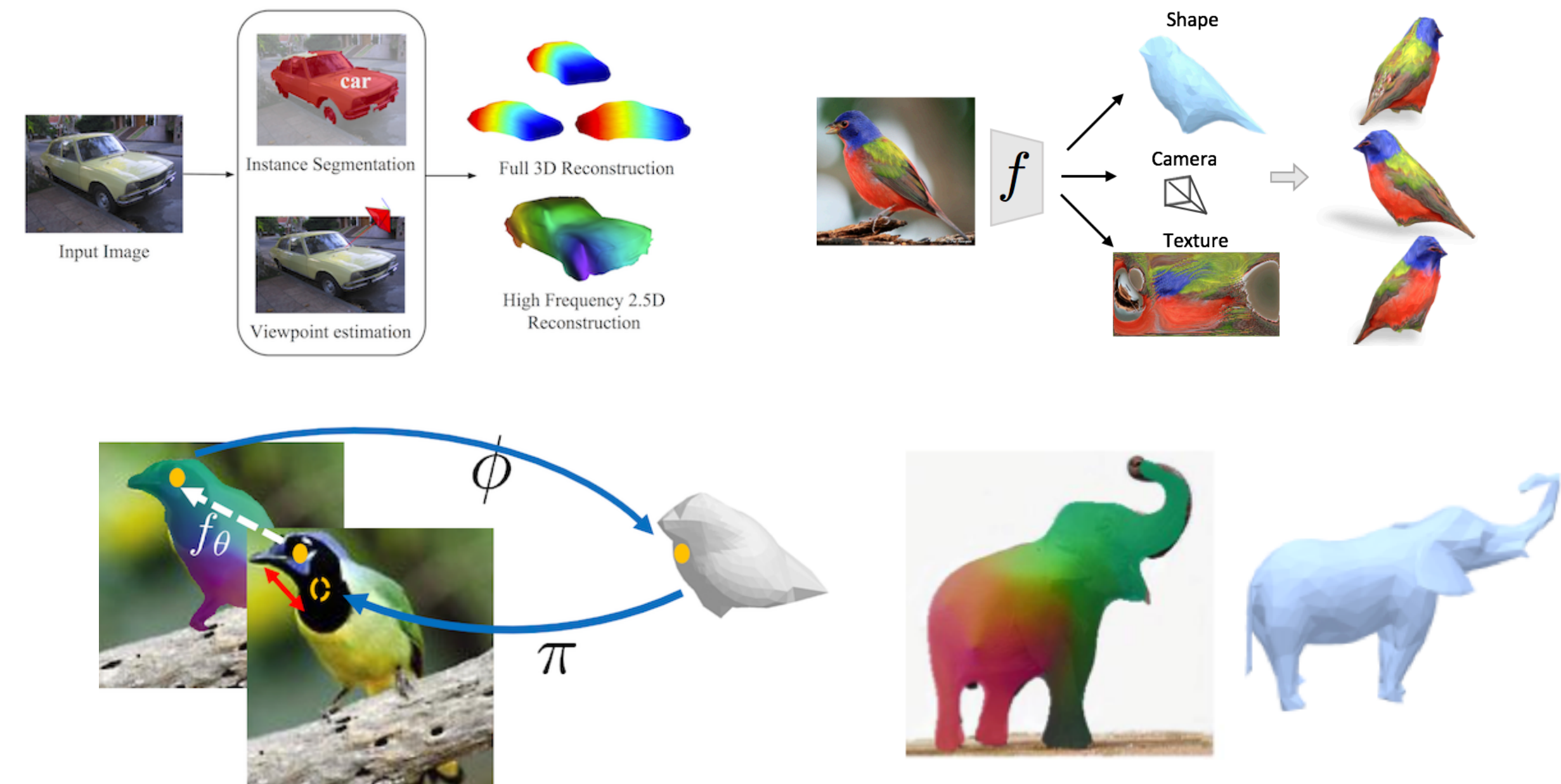
$$\mathbf{K}_{2 \times 2} = \mathbf{I}$$

**Scaled
Orthographic:**

$$\mathbf{K}_{2 \times 2} = s\mathbf{I}$$

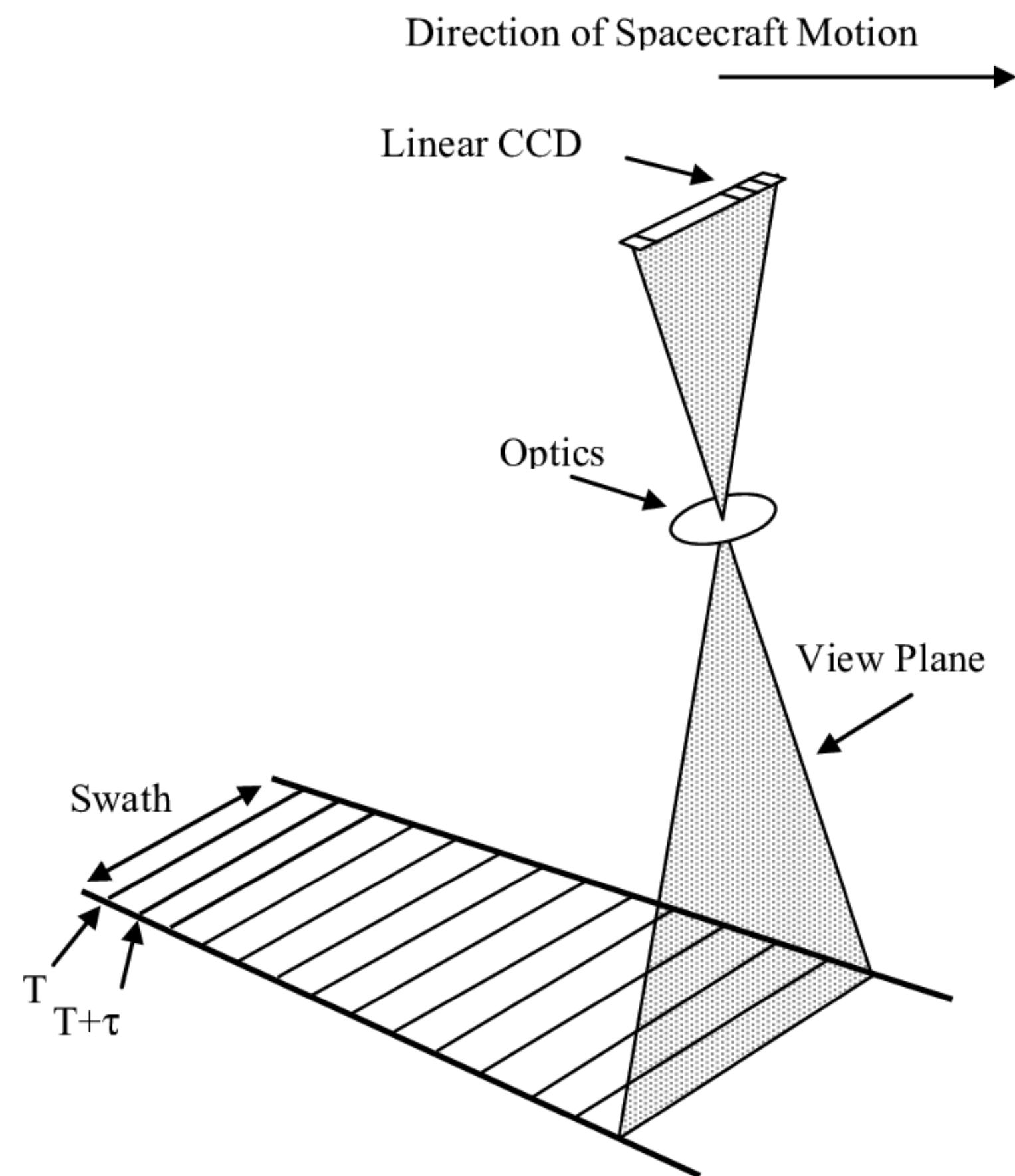
Weak Perspective:

$$\mathbf{K}_{2 \times 2} = \begin{bmatrix} s_1 & 0 \\ 0 & s_2 \end{bmatrix}$$

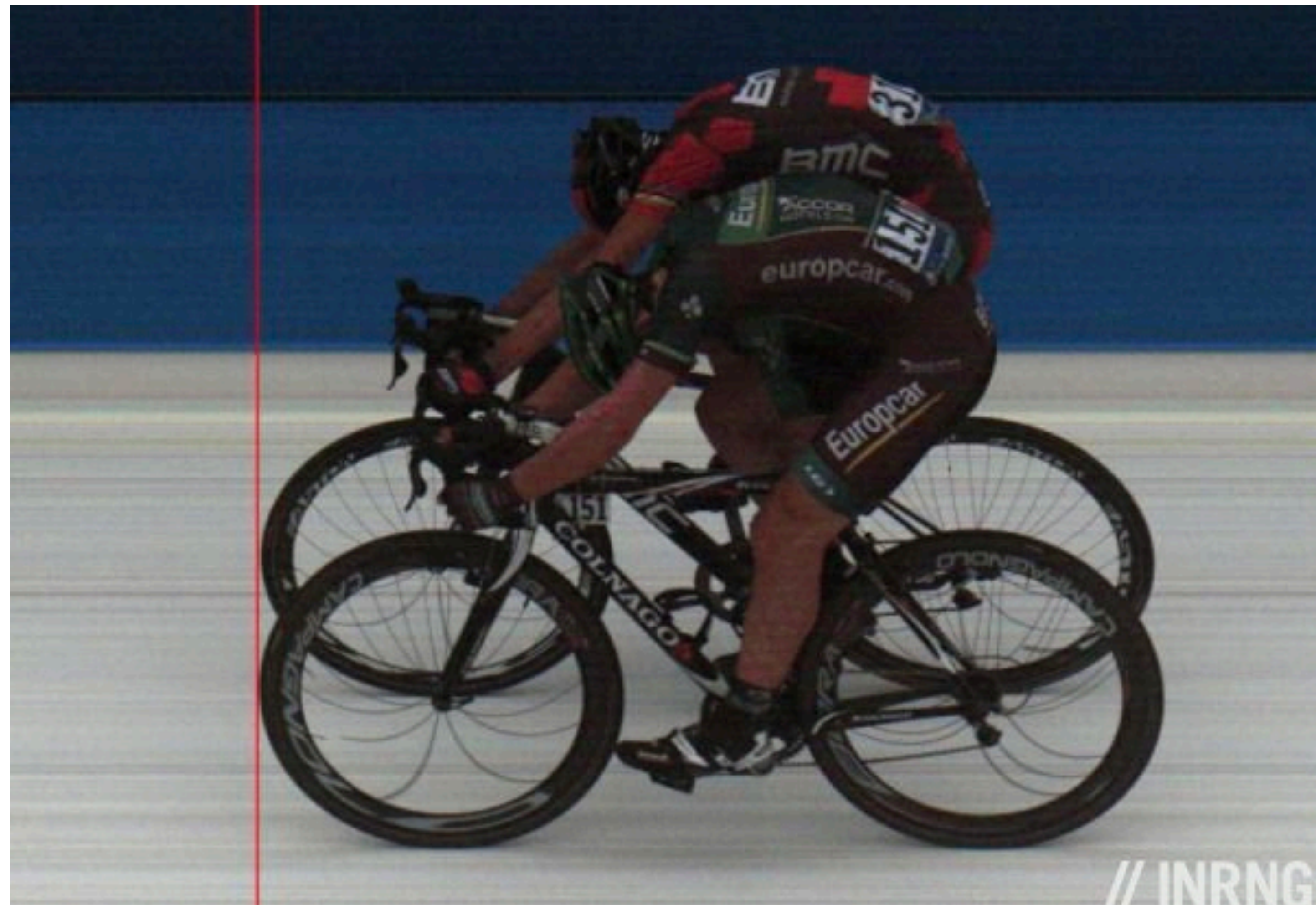


Widely used camera parametrization when learning 3D prediction from images!

Aside: Pushbroom Cameras

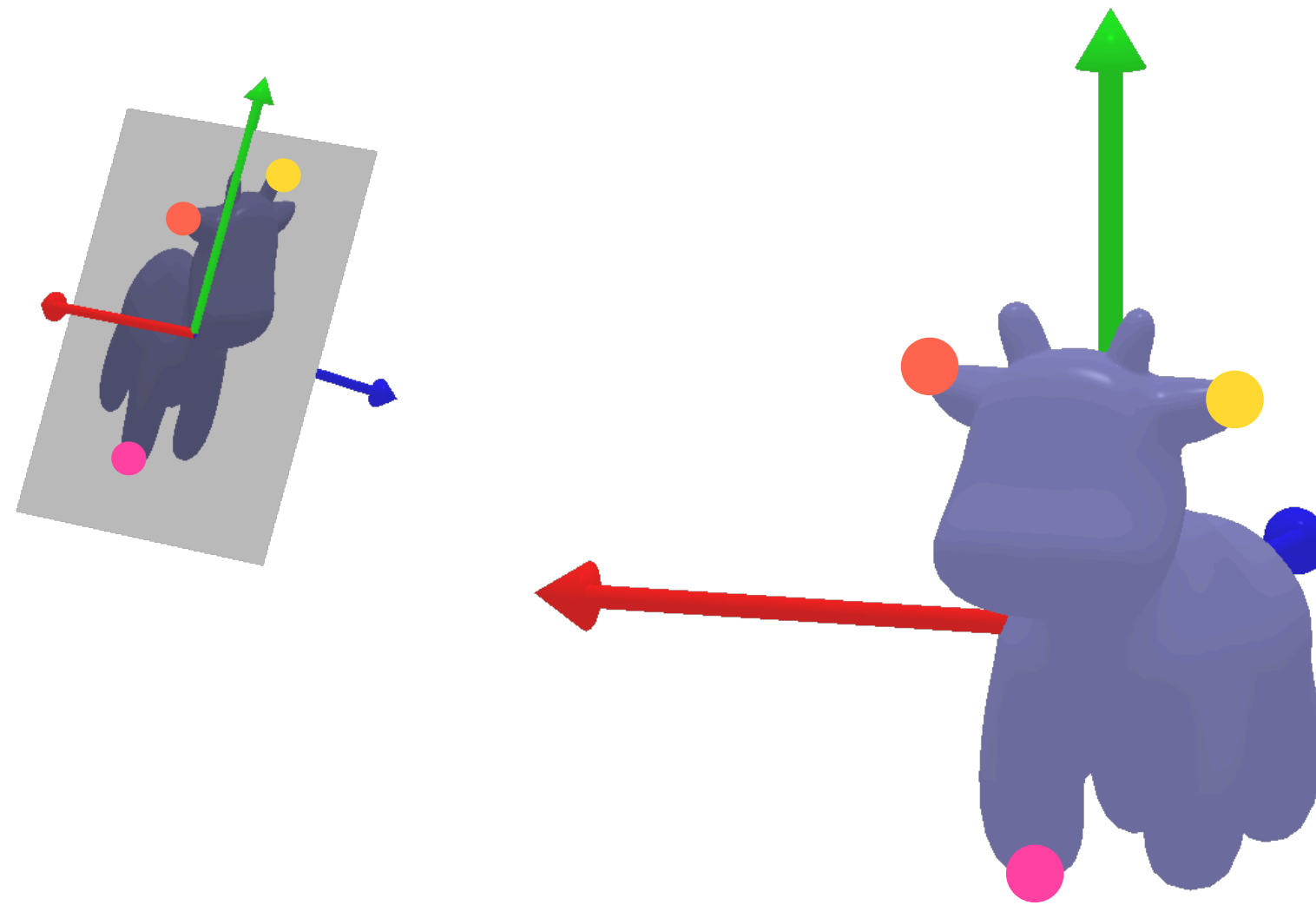


$$\mathbf{PX} = \begin{pmatrix} x, y, w \end{pmatrix}^T$$
$$\begin{pmatrix} x, y/w \end{pmatrix}^T$$



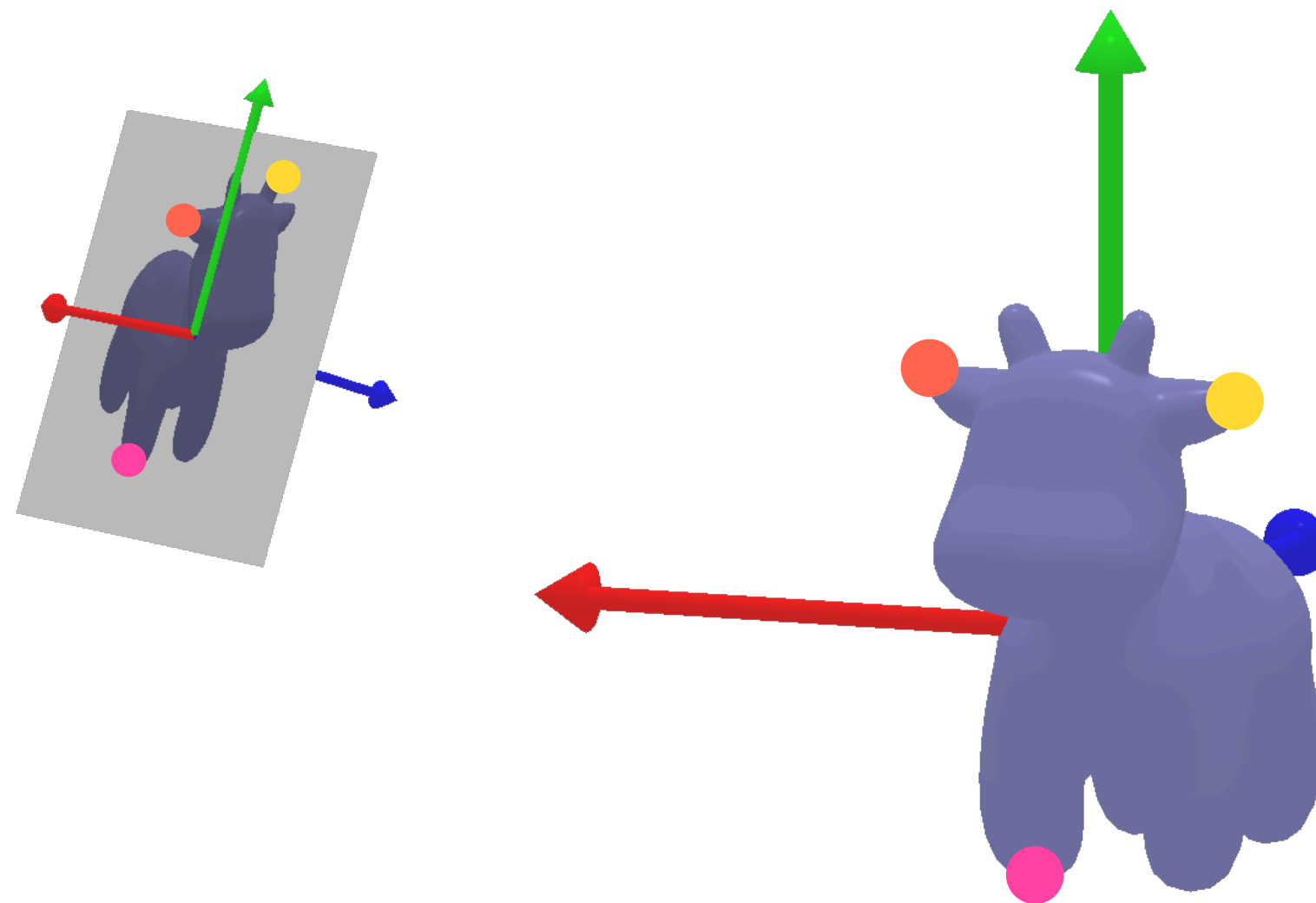
Determining the Camera Matrix

$$\mathbf{P} = \mathbf{K}R \begin{bmatrix} I & -\tilde{\mathbf{C}} \end{bmatrix} = \begin{bmatrix} \mathbf{M} & \mathbf{p}_4 \end{bmatrix}$$



Determining the Camera Matrix

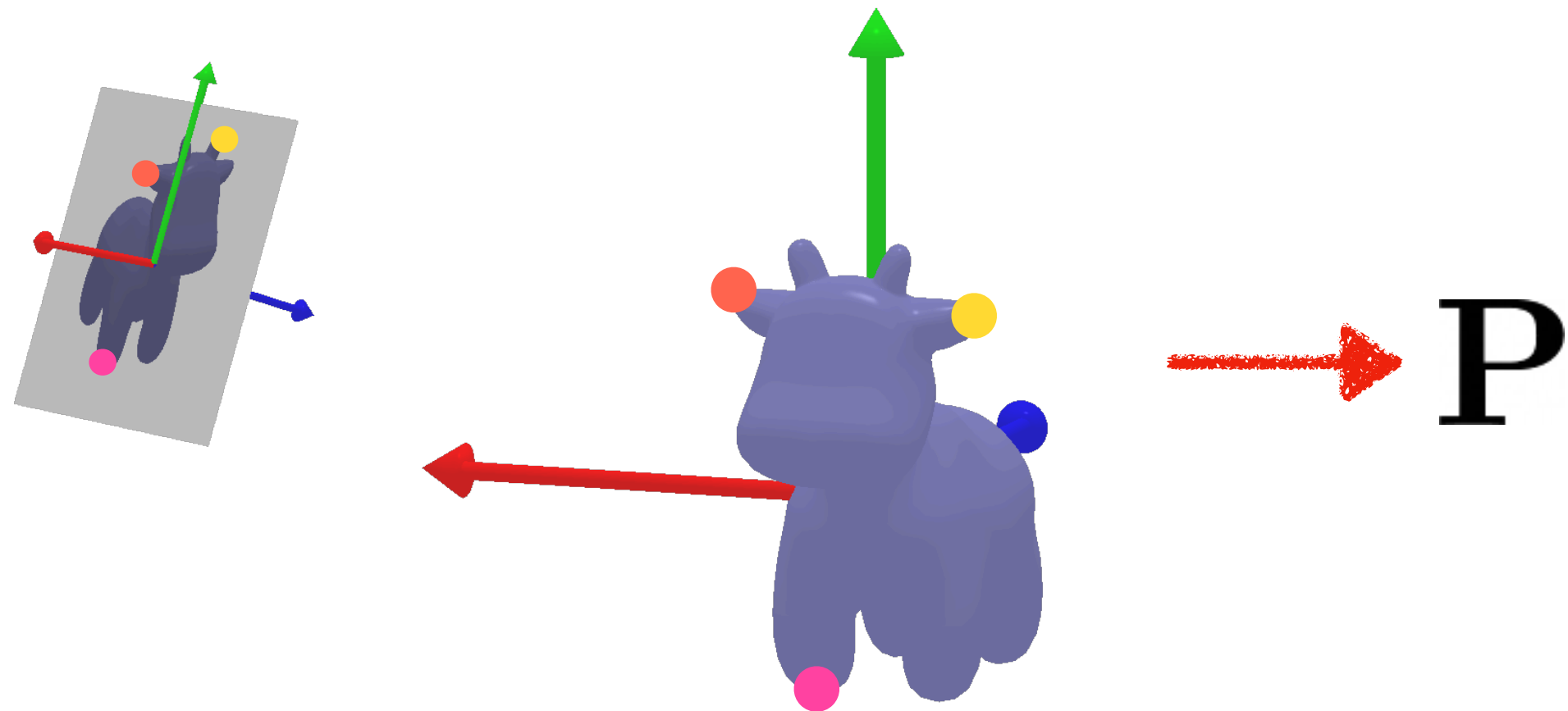
$$\mathbf{P} = \mathbf{K}R \begin{bmatrix} I & -\tilde{\mathbf{C}} \end{bmatrix} = \begin{bmatrix} \mathbf{M} & \mathbf{p}_4 \end{bmatrix}$$



Correspondences between known 3D
points and their projections

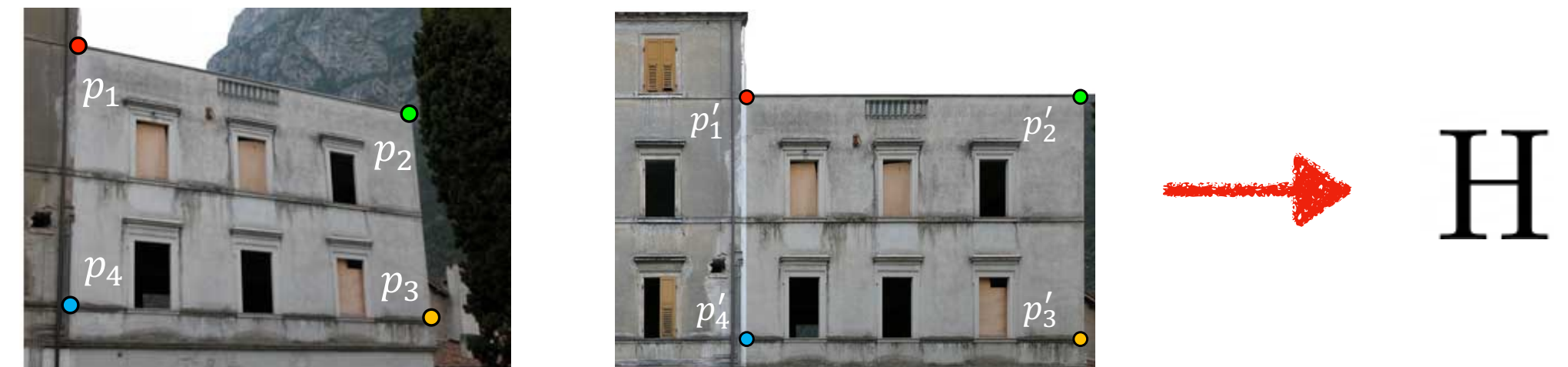
$$\mathbf{X}_i \rightarrow \mathbf{x}_i$$

Determining the Camera Matrix



$$\mathbf{P}\mathbf{X}_i = \mathbf{x}_i$$

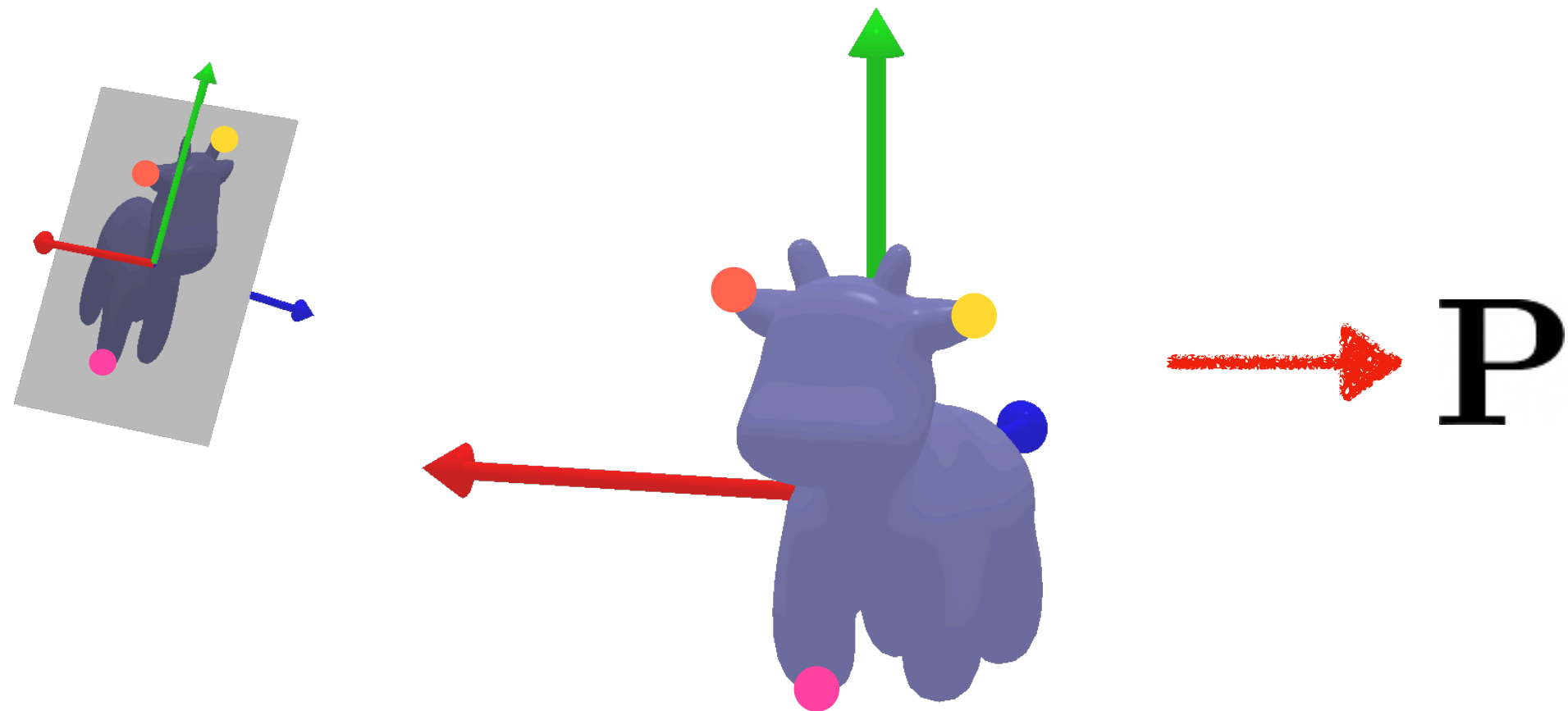
$$\mathbf{P}\mathbf{X}_i \times \mathbf{x}_i = \mathbf{0}$$



$$\mathbf{H}\mathbf{x}_i = \mathbf{x}'_i$$

$$\mathbf{H}\mathbf{x}_i \times \mathbf{x}'_i = \mathbf{0}$$

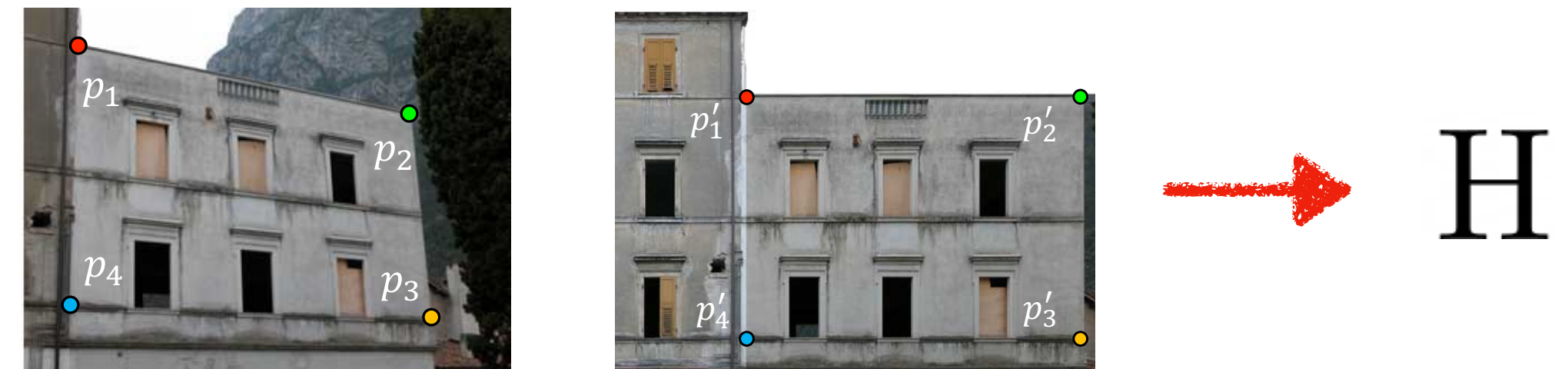
Determining the Camera Matrix



$$\mathbf{P}\mathbf{X}_i \times \mathbf{x}_i = \mathbf{0}$$

$$\mathbf{P} = \begin{bmatrix} \mathbf{p}^{1\top} \\ \mathbf{p}^{2\top} \\ \mathbf{p}^{3\top} \end{bmatrix} \quad \mathbf{p} \equiv \begin{bmatrix} \mathbf{p}_1 \\ \mathbf{p}_2 \\ \mathbf{p}_3 \end{bmatrix}$$

$$[\mathbf{x}]_{\times}^{\top} \begin{bmatrix} \mathbf{X}^{\top} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{X}^{\top} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{X}^{\top} \end{bmatrix} \mathbf{p} = \mathbf{0}$$

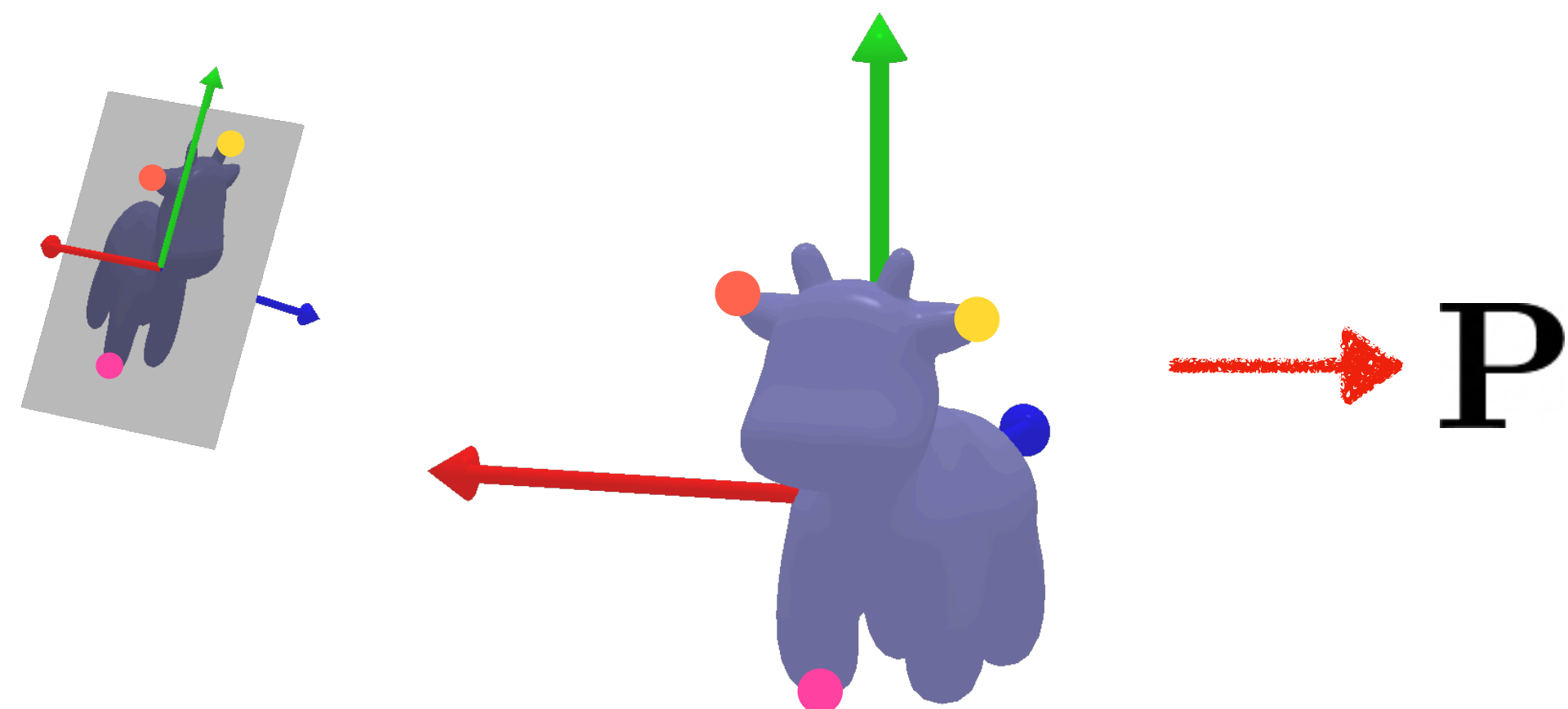


$$H\mathbf{x}_i \times \mathbf{x}'_i = \mathbf{0}$$

$$H = \begin{bmatrix} \mathbf{h}^{1\top} \\ \mathbf{h}^{2\top} \\ \mathbf{h}^{3\top} \end{bmatrix} \quad \mathbf{h} = \begin{bmatrix} \mathbf{h}^1 \\ \mathbf{h}^2 \\ \mathbf{h}^3 \end{bmatrix}$$

$$[\mathbf{x}']_{\times}^{\top} \begin{bmatrix} \mathbf{x}^{\top} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{x}^{\top} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{x}^{\top} \end{bmatrix} \mathbf{h} = \mathbf{0}$$

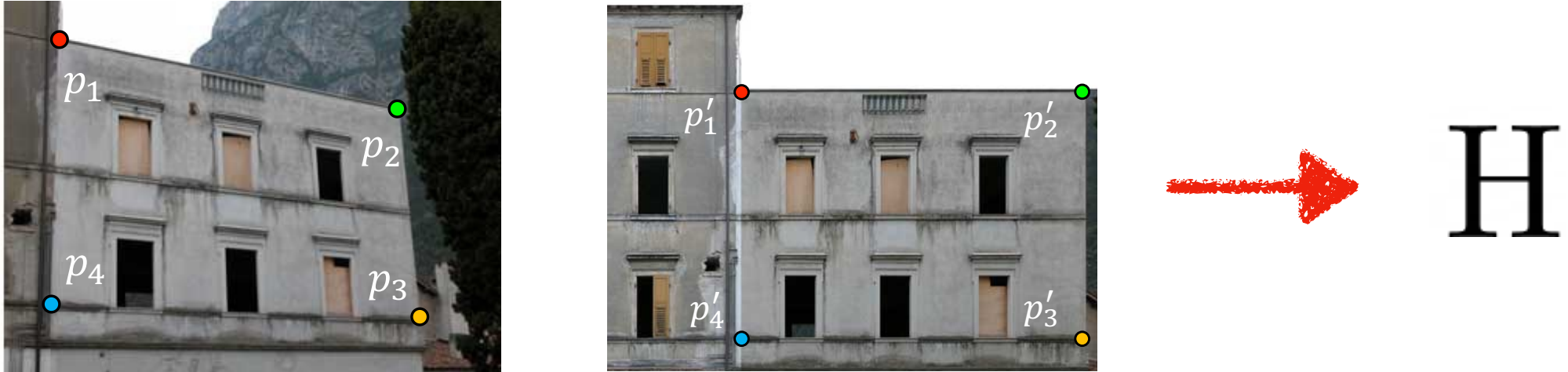
Determining the Camera Matrix



$$\mathbf{P}\mathbf{X}_i \times \mathbf{x}_i = \mathbf{0}$$

$$\begin{bmatrix} \mathbf{0} & -w\mathbf{X}^\top & y\mathbf{X}^\top \\ w\mathbf{X}^\top & \mathbf{0} & -x\mathbf{X}^\top \\ -y\mathbf{X}^\top & x\mathbf{X}^\top & \mathbf{0} \end{bmatrix} \underset{12 \times 1}{\mathbf{p}} = \mathbf{0}$$

3×12

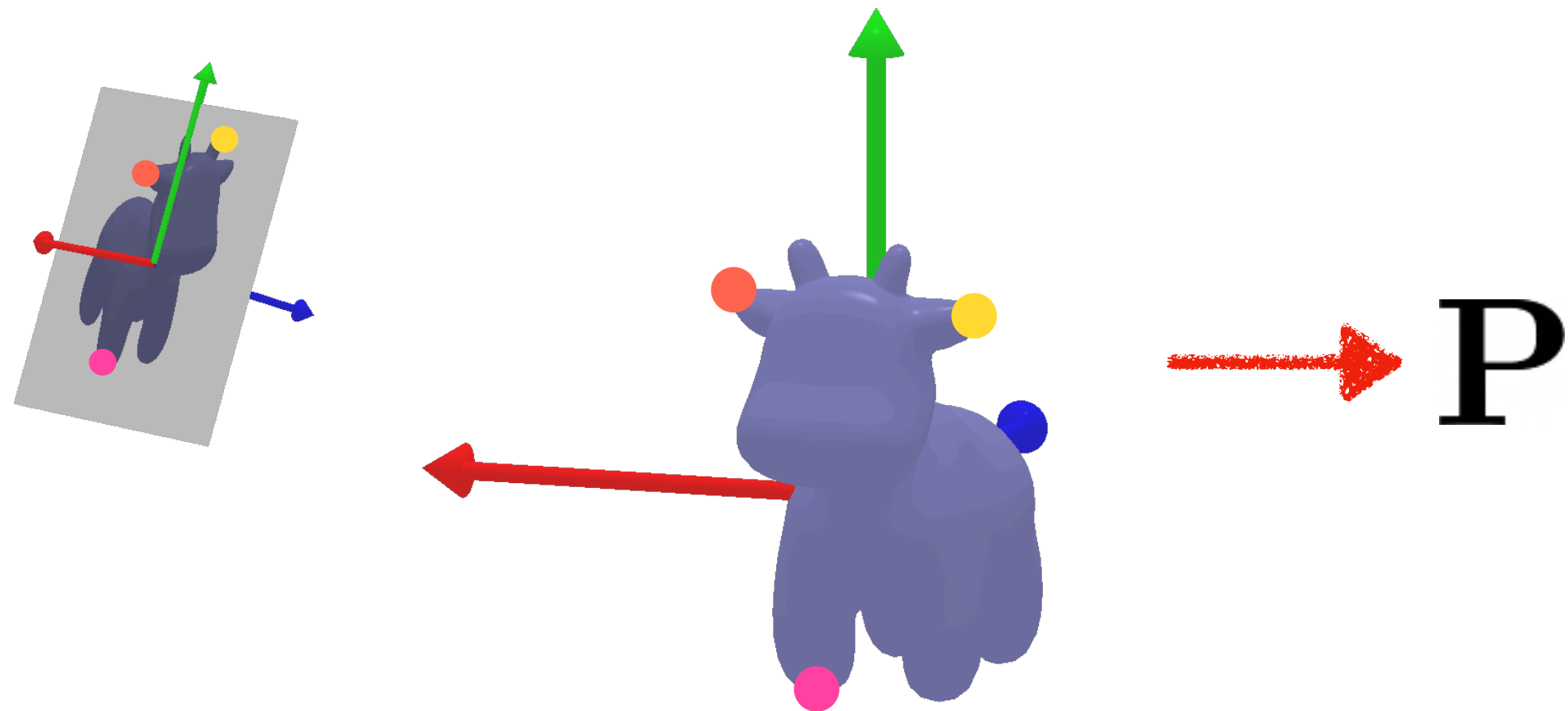


$$H\mathbf{x}_i \times \mathbf{x}'_i = \mathbf{0}$$

$$\begin{bmatrix} \mathbf{0} & -w'\mathbf{x}^\top & y'\mathbf{x}^\top \\ w'\mathbf{x}^\top & \mathbf{0} & -x'\mathbf{x}^\top \\ -y'\mathbf{x}^\top & x'\mathbf{x}^\top & \mathbf{0} \end{bmatrix} \underset{9 \times 1}{\mathbf{h}} = \mathbf{0}$$

3×9

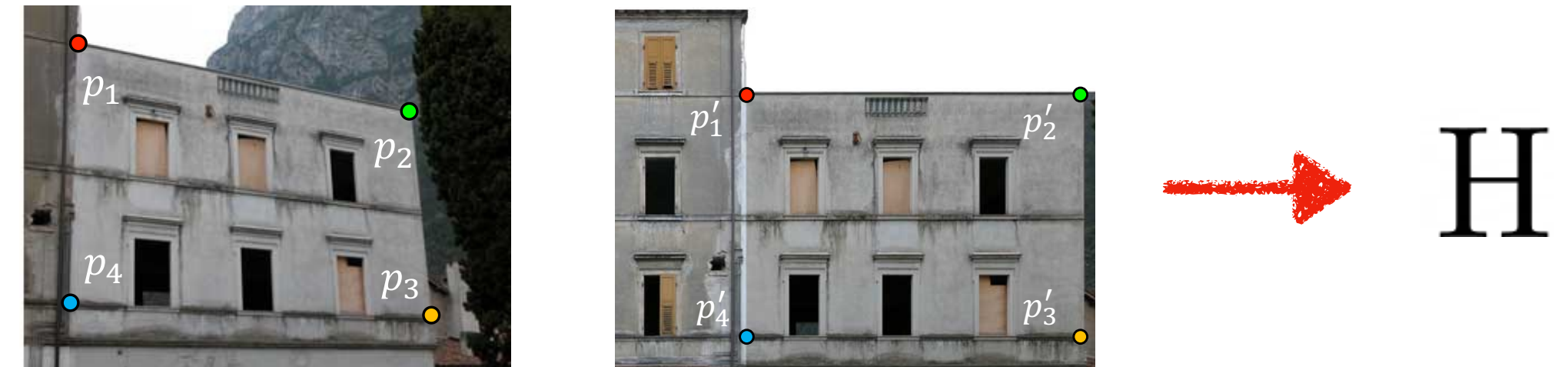
Determining the Camera Matrix



$$\mathbf{P}\mathbf{X}_i \times \mathbf{x}_i = \mathbf{0}$$

$$\min_{\mathbf{p}} \|\mathbf{A}\mathbf{p}\|$$

$$\|\mathbf{p}\| = 1$$



$$\mathbf{H}\mathbf{x}_i \times \mathbf{x}'_i = \mathbf{0}$$

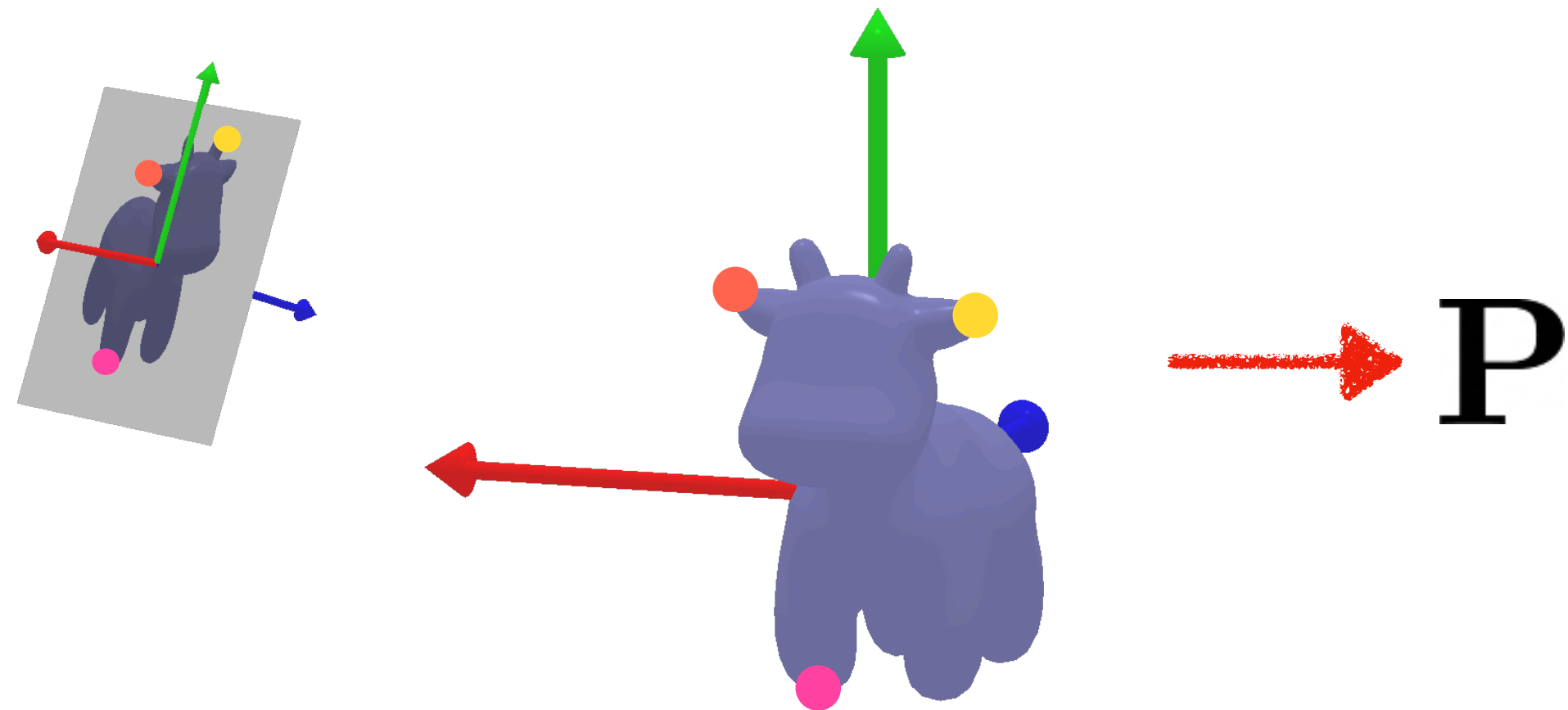
$$\min_{\mathbf{h}} \|\mathbf{A}\mathbf{h}\|$$

$$\|\mathbf{h}\| = 1$$

Other (geometric) objectives also apply!

Use DLT to initialize + iterative optimization

Determining the Camera Matrix

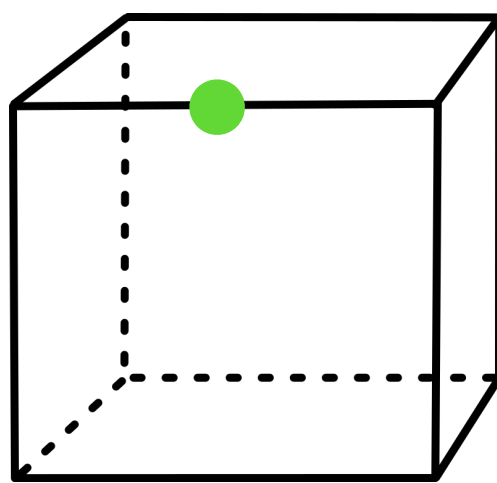


$$\mathbf{P}\mathbf{X}_i \times \mathbf{x}_i = \mathbf{0}$$

$$\min_{\mathbf{p}} \|\mathbf{A}\mathbf{p}\| \quad \|\mathbf{p}\| = 1$$

Q: Other annotations beyond point correspondences?

3D Point $\mathbf{X}$



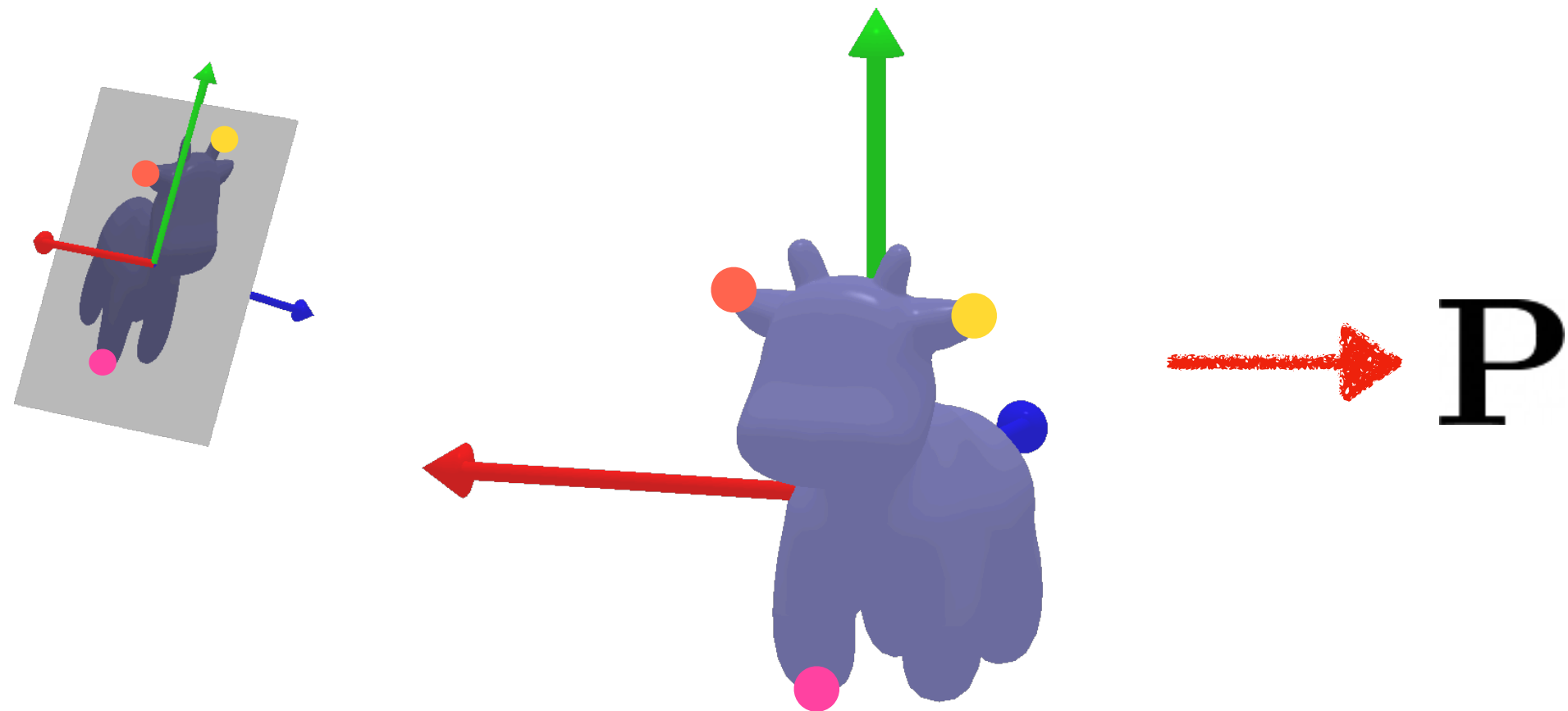
2D line $\mathbf{l}$



$$\mathbf{l}^\top \mathbf{P}\mathbf{X} = 0$$

Only yields one constraint on $\mathbf{P}$

Determining the Camera Matrix



$$\mathbf{P}\mathbf{X}_i \times \mathbf{x}_i = \mathbf{0}$$

$$\min_{\mathbf{p}} \|\mathbf{A}\mathbf{p}\| \quad \|\mathbf{p}\| = 1$$

Q: What if some additional information on $\mathbf{P}$ is known?

*Not always reducible to analytical optimization
(e.g. known $\mathbf{K}$)*

Known camera location

Known extrinsics

$$\mathbf{K}\mathbf{R} \begin{bmatrix} \mathbf{I} & -\tilde{\mathbf{C}} \end{bmatrix} \mathbf{X}_i \times \mathbf{x}_i = \mathbf{0}$$

Square pixels

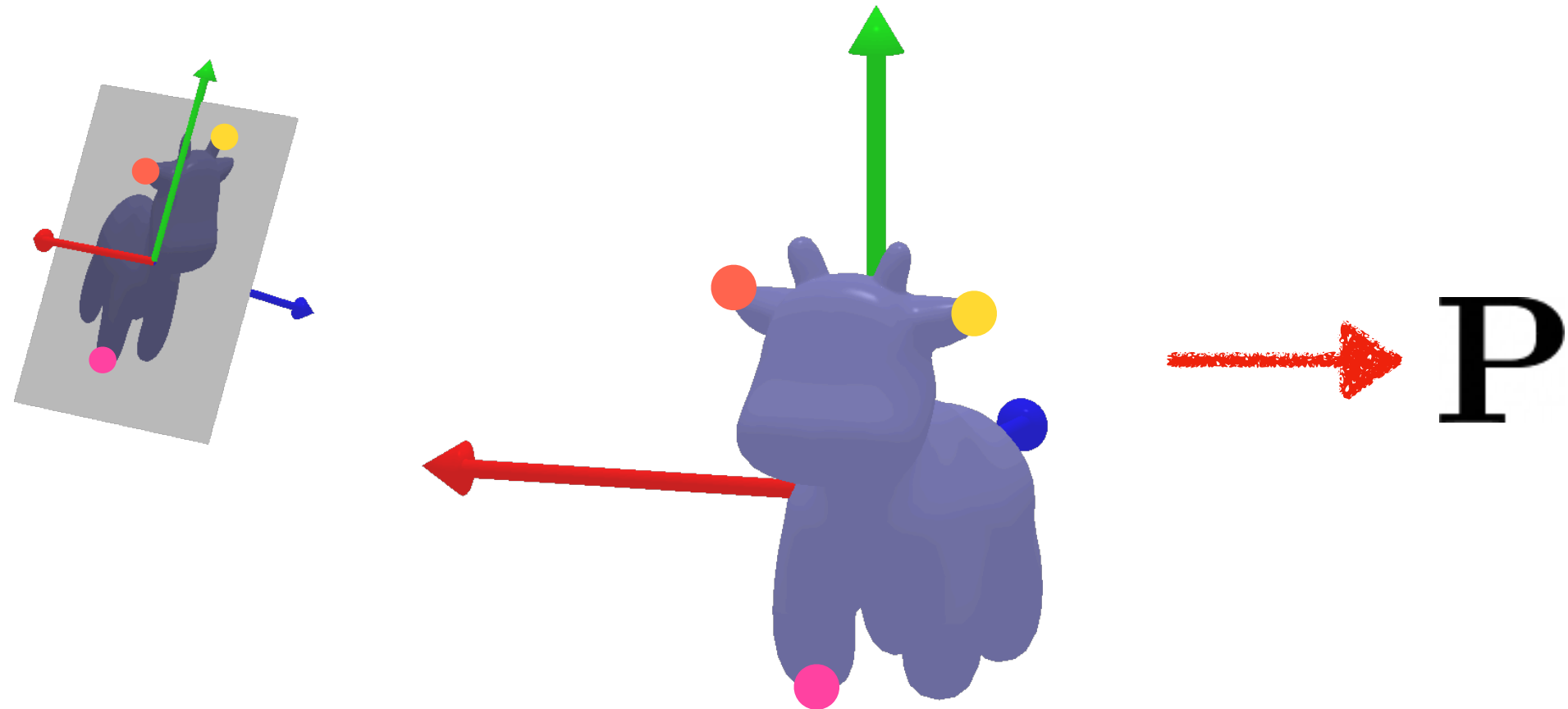
Known camera orientation

$$\min_{\mathbf{p}} \|\mathbf{A}\mathbf{p}\|$$

$$\mathbf{R} \in SE(3)$$

Known intrinsics

Determining the Camera Matrix



$$\mathbf{P} \mathbf{X}_i \times \mathbf{x}_i = \mathbf{0}$$

$$\min_{\mathbf{p}} \|\mathbf{A} \mathbf{p}\| \quad \|\mathbf{p}\| = 1$$

Q: What if some additional information on P is known?

$$\mathbf{m}^{3T} = \mathbf{0}$$

Affine Camera

Known camera location

But sometimes possible!

Known extrinsics

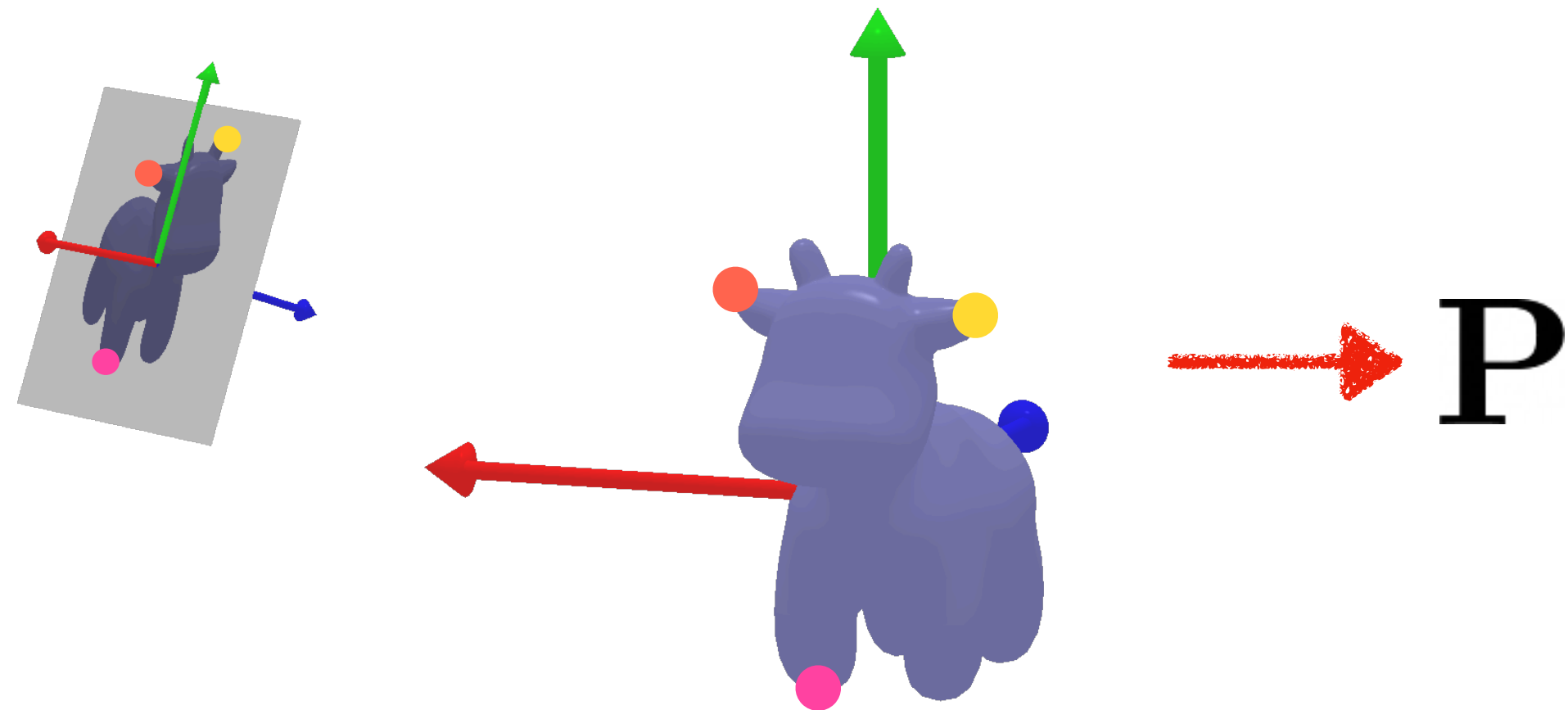
$$\mathbf{K} \bar{\mathbf{x}}_i \times \mathbf{x}_i = \mathbf{0}$$

$$\bar{\mathbf{x}}_i = \begin{bmatrix} \mathbf{R} & \mathbf{t} \end{bmatrix} \mathbf{X}_i$$

$$\mathbf{P} \begin{bmatrix} \tilde{\mathbf{C}} \\ 1 \end{bmatrix} = \mathbf{0}$$

Q: How many point correspondences required in each case?

Determining the Camera Matrix



$$\mathbf{P}\mathbf{X}_i \times \mathbf{x}_i = \mathbf{0}$$

$$\min_{\mathbf{p}} \|\mathbf{A}\mathbf{p}\| \quad \|\mathbf{p}\| = 1$$

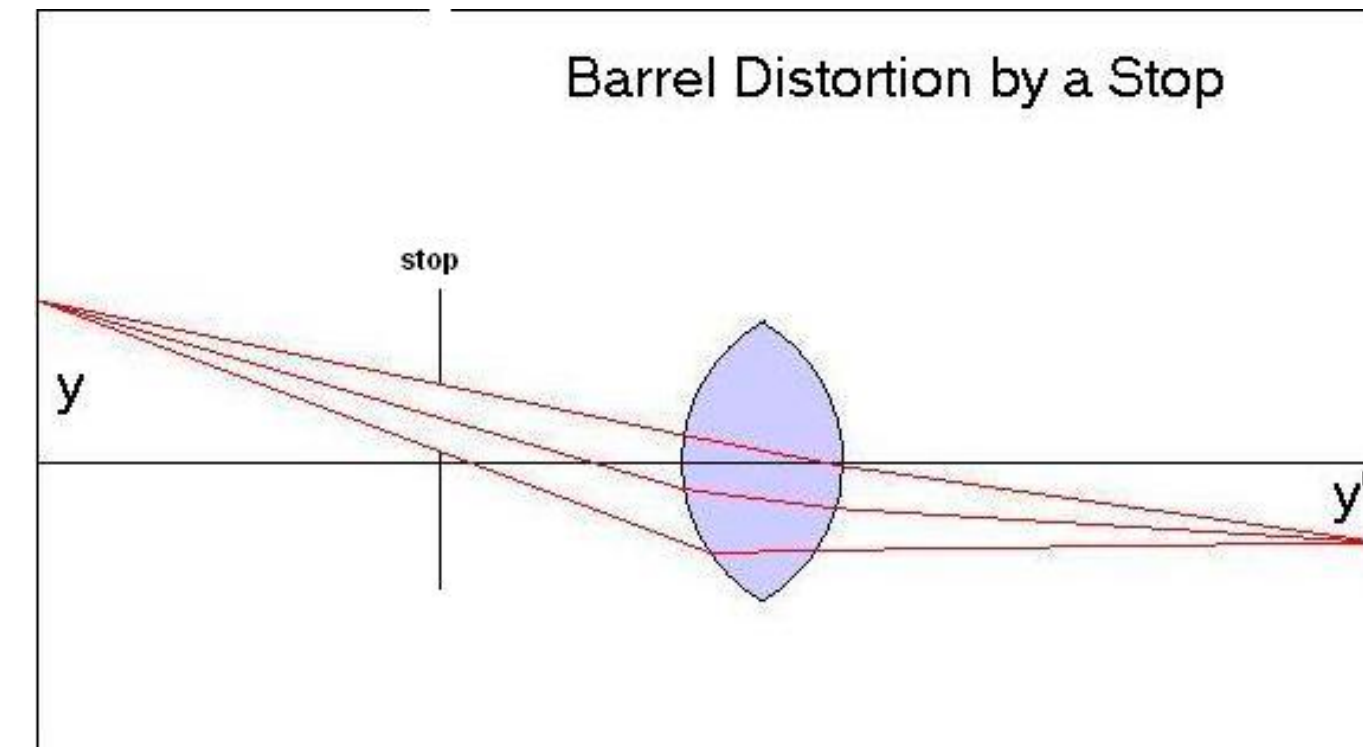
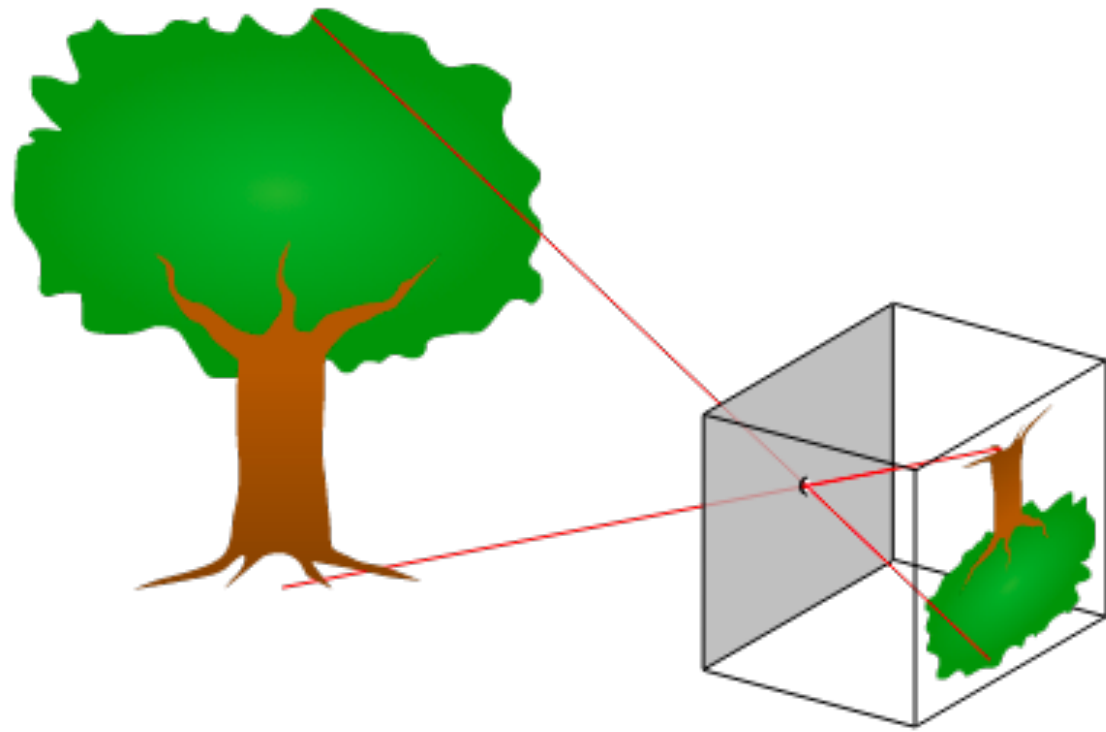
Q: What if some additional information on P is known?

Known K , estimate R , $\mathbf{t}$

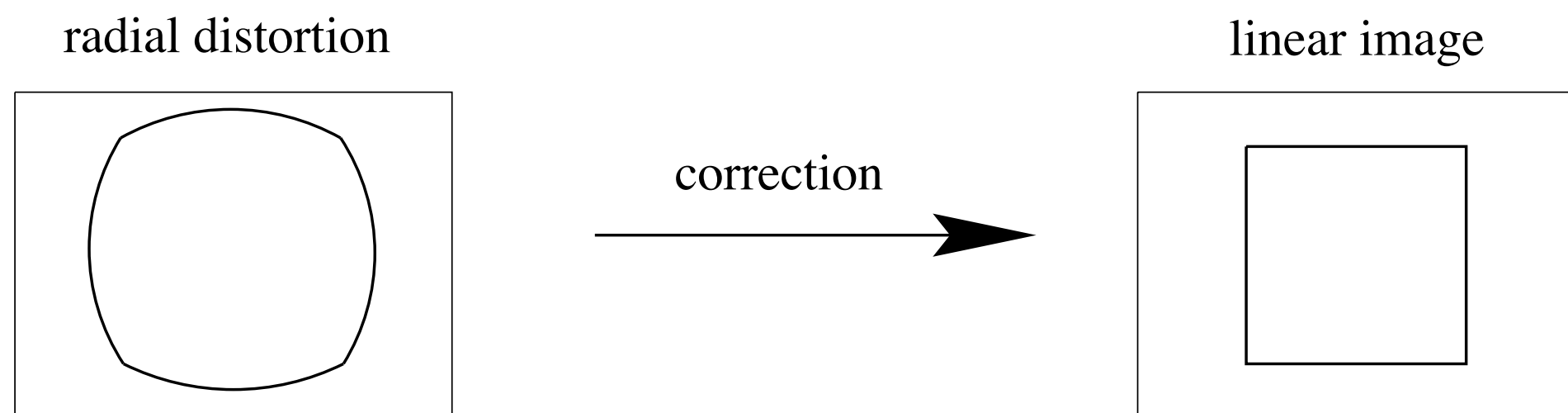
Q: How many
correspondences required?

3 correspondences suffice, but yields 4 solutions all of
which satisfy the constraints

Aside: Radial Distortion



Real cameras are not pinhole



Step 1: Assume a distortion model relating 'ideal' and observed points

$$(\tilde{x}, \tilde{y}, 1)^T = \mathbf{K} \begin{bmatrix} I & \mathbf{0} \end{bmatrix} \mathbf{X}_{cam}$$

Step 2: Optimize coefficients using constraints e.g. image of lines is straight

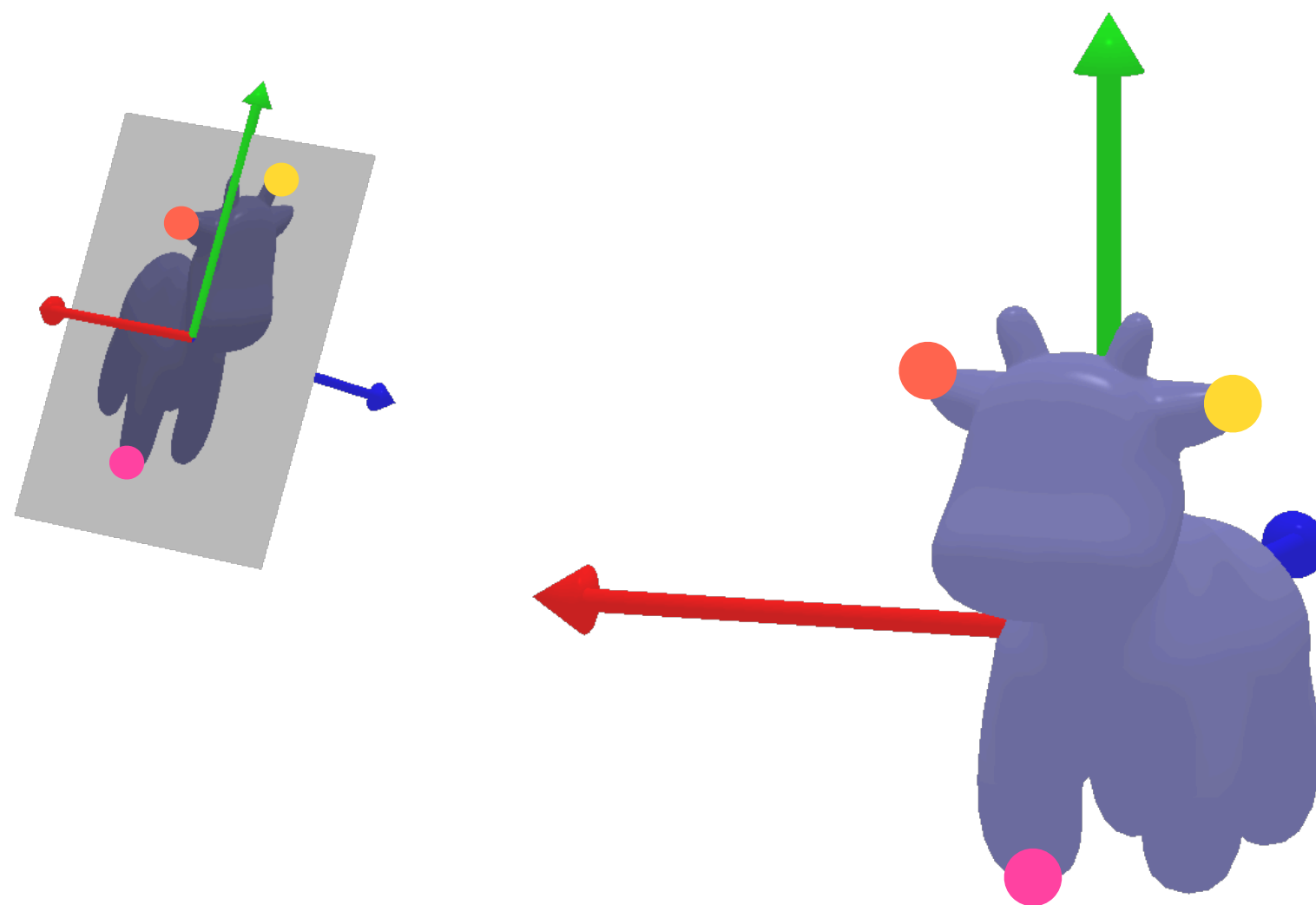
$$\begin{pmatrix} x_d \\ y_d \end{pmatrix} = L(\tilde{r}) \begin{pmatrix} \tilde{x} \\ \tilde{y} \end{pmatrix}$$

Determining the Camera Matrix

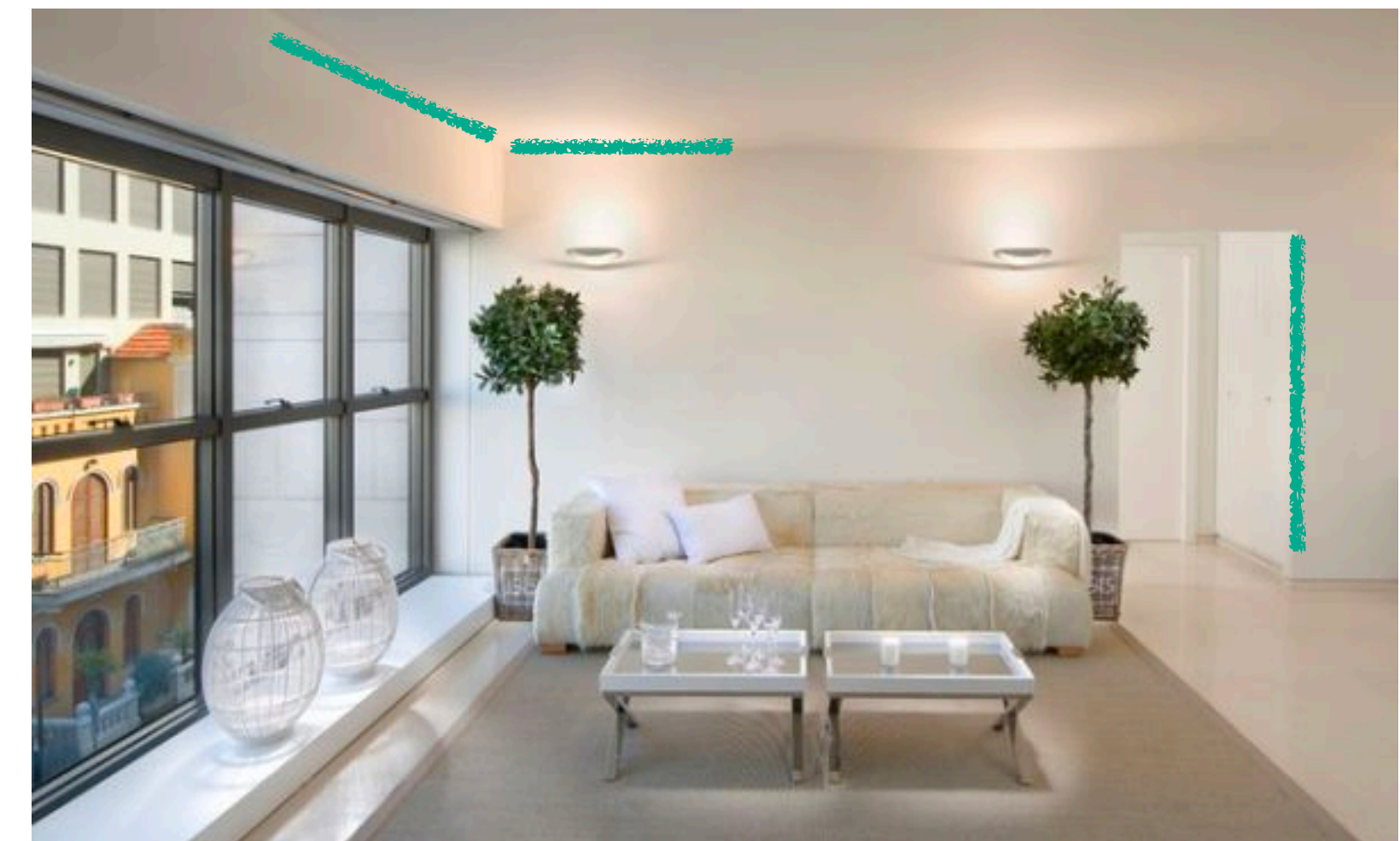
$$\mathbf{P} = \mathbf{K}R \begin{bmatrix} I & -\tilde{\mathbf{C}} \end{bmatrix}$$

$$\mathbf{P} = \mathbf{K}R \begin{bmatrix} I & -\tilde{\mathbf{C}} \end{bmatrix}$$

Possible to recover intrinsics from annotation!



Can recover both, world to camera,
and camera to pixel transforms



What if the underlying 3D model is unknown?

next lecture!

Questions?